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A COMPOSITIONAL AND DIAGRAMMATIC SYNTAX FOR PARAMETRIC LINEAR
PROGRAMMING

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Dissertação de Mestrado apresentada ao Programa de
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e Instituto Tércio Pacitti de Aplicações e Pesquisas
Computacionais, Universidade Federal do Rio de Janeiro,
como requisito parcial à obtenção do título de Mestre
em Informática.

Orientador: Prof. João Paixão, D.Sc.

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RESUMO

Esta dissertação apresenta um percurso categórico pela teoria da Álgebra Gráfica em direção a um cálculo composicional para problemas de otimização. Investiga-se a relação intrínseca entre composição/eliminação de variáveis e o próprio processo de otimização, mostrando como, sob determinadas condições concretas, a otimização emerge da estrutura interna de categorias enriquecidas sobre quantales. O resultado central deste trabalho é a apresentação de uma teoria monoidal completa para programação linear paramétrica no lado direito.

O desenvolvimento parte das noções necessárias de quantales e relações, bem como de sua extensão a espaços métricos por meio do quantale de Lawvere. Em seguida, retoma-se e explora-se a álgebra diagramática já existente para poliedros, **GPA**, fornecendo uma demonstração completa de sua correção e completude. Por fim, essa teoria é estendida por meio de uma abordagem simples e elegante, com a adição de um único gerador e três axiomas, elevando a linguagem de uma representação meramente de poliedros para uma álgebra mais expressiva, capaz de denotar Programas Lineares Paramétricos.

O cálculo diagramático desenvolvido aqui representa um passo em direção ao objetivo mais amplo de fornecer uma linguagem mais intuitiva e prática para o raciocínio computacional e matemático em otimização. Nesse contexto, detalhes técnicos podem tornar-se difíceis de administrar, particularmente ao se raciocinar sobre sistemas complexos. Os diagramas de cordas ajudam a ocultar muitos desses pormenores por trás de uma sintaxe gráfica compacta, tornando a estrutura abstrata subjacente e suas propriedades fundamentais mais diretamente acessíveis à manipulação e ao raciocínio.

Palavras-chave: teoria das categorias; diagramas de corda; programação linear; otimização convexa; categorias monoidais simétricas; relações ponderadas; quantais; álgebra gráfica; formas normais; completude.

ABSTRACT

This dissertation presents a categorical roadmap through the theory of Graphical Algebra towards a compositional calculus for Optimization problems. It investigates the intrinsic relationship between composition /variable elimination and that of optimization itself. Showing how under concrete settings, optimization emerges from the internal structure of quantale-categories. The central result of this work is the presentation of a complete monoidal theory for right-hand side parametric linear programming.

It builds up from the necessary notions of quantales and relations and it's extension towards metric-spaces through the Lawvere-Quantale. Then it recalls and explore the already existing diagrammatic algebra for polyhedrals **GPA** providing the full demonstration of it's soundness and completeness. In the end, it increments this existing theory through a elegant and simple approach, adding one single generator and three axioms, lifting the language from simply expressing polyhedrals to a more powerfull algebra capable of denoting Parametric Linear Programs.

The diagrammatic calculus developed here represents a step toward the broader goal of providing a more intuitive and practical language for computational and mathematical reasoning in optimization. In this setting, technical details can become difficult to navigate, particularly when reasoning about complex systems. String diagrams help by hiding many of these particulars behind a compact graphical syntax, thereby making the underlying abstract structure and core properties more directly accessible to manipulation and reasoning.

Keywords: category theory; string diagrams; linear programming; convex optimization; symmetric monoidal categories; weighted relations; quantales; graphical algebra; normal forms; completeness.

STATEMENT ON THE USE OF ARTIFICIAL INTELLIGENCE

The use of AI in this paper is described in a few different ways. When it comes to the writing of the thesis it was restricted to the following use:

- Tikz Writing: Convert tables and diagrams from handwritten images to tikz syntax.
- Writing review: AI searched for grammatical inconsistencies, typos, open-ended and poorly-formed sentences.
- Review the mathematical argument: The AI was instructed to look for loops or failed arguments across mathematical proofs and derivations.
- Loop Review: Read and evaluate the work multiple times and provide critics and feedback about the overall writing.
- Latex Formatting: AI was used to adapt the paper to the necessary standards demanded by a dissertation. This refers only to document structure(font size, title format, citations etc) not the writing content.

AI most relevant effort came in into the appendix for helping writing down a long list of well stablished mathematical definitions to provide the reader the necessary background. Text outside definitions are not a product of AI use, AI only wrote the definitions itself. The main body of the work (from introduction to conclusion) is almost entirely written by hand having AI only as a default reviewer.

During the development and research of the thesis, AI was used as a learning gate and searching mechanism to discuss ideas, look for references, investigate topics and understand papers and works from others.

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1 INTRODUCTION

Mathematics is permeated by this notion of what one would call an “ideal” object. This notion of “ideal” often refers to the simplest object that meets a specific criterion, such as being the smallest element in a set, or the one through which all others factor uniquely. This particular description of “ideal” reveals an interesting interconnection between categories and optimization [Kad26]. Many of the constructions mathematics singles out as canonical are optimal in this same sense: they are the best object of a collection, with respect to an ordering or universal property carried by that collection. This thesis aims to explore this idea and provide a formal and intuitive view on such topics.

Consider for instance the familiar transitivity condition for a relation $R \subseteq X \times X$:

$$(\exists y \in X : xRy \wedge yRz) \implies xRz, \quad x, z \in X.$$

Here the premise demands the existence of an intermediate object, namely y . Nothing forces y to be unique; in fact, it is often the case that we have a set of such objects. The connection with an “ideal” object can be made explicit if we change the settings of the problem. Consider the indicator function representation of R in the sense of convex analysis [Roc70], $R : X \times X \rightarrow \{0, \infty\}$, defined by

$$R(x, y) = \begin{cases} 0, & \text{if } xRy, \\ \infty, & \text{otherwise.} \end{cases}$$

Then the above is the same as:

$$\inf_{y \in X} (R(x, y) + R(y, z)) = 0 \implies R(x, z) = 0,$$

where $\inf \emptyset = \infty$. Since the values are either zero or infinity, this is also equivalent to

$$\inf_{y \in X} (R(x, y) + R(y, z)) \geq R(x, z)$$

The resemblance is not mere coincidence: these are particular cases of a more general construction using quantales [Law73]. More generally, let Q be a unital quantale: a complete lattice equipped with an associative operation $\otimes$, with unit e , that preserves arbitrary joins in each argument. By interpreting $\exists$ and $\inf$ through the lattice join operator $\bigvee$, and $\wedge$ and $+$ through the monoidal operator $\otimes$, we obtain, for a Q -valued relation $R : X \times X \rightarrow Q$, the generalized transitivity condition

$$\bigvee_{y \in X} R(x, y) \otimes R(y, z) \leq_Q R(x, z).$$

Notice that for the cost quantale $([0, \infty], \geq, +)$ we have $a \leq_Q b \iff a \geq b$, so this replacement recovers exactly the same expression as above.

26 However, there is still more to it: as one may notice, the commonalities between these
 27 equations go beyond the simply visual structure; they actually share an *algebraic* structure.
 28 Explicitly, all the expressions on the left of the abstract inequality perform the very important
 29 role of *variable elimination*. The two terms share the same middle variable y , and taking the join
 30 over y produces a new expression depending only on x and z .

31 A join, however, already has an optimization characterization: for $A \subseteq Q$, $\bigvee A$ is the least of
 32 the upper bounds of A , the “ideal” upper bound inside the order of Q . Here the collection being
 33 aggregated is

$$A_{x,z} = \{R(x, y) \otimes R(y, z) : y \in X\}.$$

34 Eliminating the intermediate variable therefore computes an optimal aggregate value. The choice
 35 of Q fixes both the order used for aggregation and the operation used to combine values. Ordinary
 36 existential quantification is thus a particular case of this order-theoretic aggregation, and its
 37 indicator function makes the connection with optimization exact.

38 This variable elimination is precisely how composition of quantale-valued relations is defined.
 39 Given $R : X \times Y \rightarrow Q$ and $S : Y \times Z \rightarrow Q$, we have

$$(S \circ R)(x, z) := \bigvee_{y \in Y} R(x, y) \otimes S(y, z).$$

40 The generalized transitivity condition above is therefore simply

$$R \circ R \leq_Q R,$$

41 with the inequality understood pointwise. In this setting, composition is variable elimination,
 42 and the join gives it the optimization interpretation described above. Sections 2.1 and 2.2 make
 43 this precise.

44 Category theory gives us an abstraction to describe such operations: it explicitly encodes
 45 the notion of composition inside its own definition. Quantale-valued relations form a category
 46 under the operation above, with identity taking value e on the diagonal and $\perp$ elsewhere. More
 47 generally, composition in a category $\mathcal{C}$ is given by

$$\circ : \mathcal{C}(Y, Z) \times \mathcal{C}(X, Y) \longrightarrow \mathcal{C}(X, Z),$$

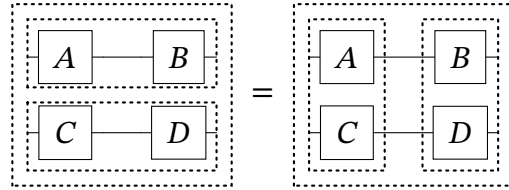
48 subject to associativity and identity laws. Its particular expression as a join over an intermediate
 49 variable uses the additional quantale-valued structure.

50 We show that, in the categories treated here, solving an optimization problem amounts to
 51 composing morphisms, and that composition carries out the variable elimination such a solution
 52 requires. Convex optimization then admits a single compositional presentation [SS24; Han+24],
 53 in which optimization is an algebraic operation rather than a technique applied from outside.
 54 That is, optimization and variable elimination are a syntactic operation, namely the composition

◦ of two objects A and B that share a common variable, or type.

This compositional view of optimization becomes especially apparent when analyzed through the language of string diagrams [JS91; Sel10; PZ23]. Originating as a rigorous graphical language for monoidal categories, string diagrams provide an intuitive yet mathematically precise syntax in which the sequential and parallel composition of morphisms correspond directly to vertical and horizontal juxtaposition of diagrams (e.g., representing computational processes) [JS91; Lan98]. Graphical languages are remarkably well suited to the case at hand. They make the semantics of the underlying mathematics explicit at the syntax level, in the sense that one doesn't need to go through atomic and subtle operations in order to express the core ideas of a theory, *but the system's main properties are explicit in the syntax operations themselves*.

For example, consider the following diagram. Because of the monoidal structure one may parse it by reading in two manners: composing the boxes in the same line and then stack them together, or stacking the boxes that are vertically aligned and then composing them sequentially, as the dotted boxes shows.



This denotes the fact that the composition and the monoidal product interact nicely, or, formally, that the operator $\otimes$ is actually an endofunctor, so $a; b \otimes c; d = a \otimes c; b \otimes d$. Observe how this entire property is underloaded by the syntax, so one don't even need to keep track of it but simply parse the diagram in any direction. This is what we mean when we say that the main properties are in the syntax itself, we don't get to invoke the definition or property to show the equality, but the very depicting (or drawing) embeds the property.

A fine example, that will be later explained, is one that uses the very syntax developed in this work. Consider the equivalence between these two Linear Programs

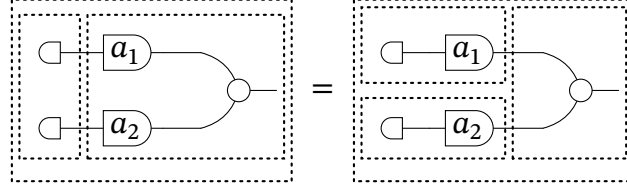
$$f_1(y) = \inf_{x_1, x_2} \{x_1 + x_2 \mid a_1 x_1 + a_2 x_2 = y\}$$

$$f_2(y) = \inf_{w_1, w_2} \{ \inf_{x_1} \{x_1 \mid a_1 x_1 = w_1\} + \inf_{x_2} \{x_2 \mid a_2 x_2 = w_2\} \mid w_1 + w_2 = y \}$$

These two functions are, in fact, the same. To see this, introduce the change of variables $w_1 = a_1 x_1$, $w_2 = a_2 x_2$.

Assuming $a_1, a_2 \neq 0$, each inner optimization problem has a unique feasible point, so substituting $w_i = a_i x_i$ yields $f_2(y) = \inf_{a_1 x_1 + a_2 x_2 = y} \{x_1 + x_2\} = f_1(y)$.

Thus, the second formulation does not change the underlying optimization problem; rather, it decomposes the original constraint through the intermediate variables w_1 and w_2 . Now look at the diagram denoting these same problems.



85 Once again, both problems are depicted as the very same diagram, being just a different order
 86 of parsing it. In this case all the variable changing was hidden by the syntax, and only the main
 87 semantics – the *problem itself* – is exposed.

88 Other areas have far off recognize the benefits of such denotation, mainly in engineering fields
 89 such as *quantum computing* [CD11] and *electronics* [BF18]. More recently, mathematicians and
 90 computer scientists have also built upon this foundation deriving a rich family of *diagrammatic*
 91 or *graphical calculi*, offering complete axiomatisations of various algebraic structures purely
 92 through equational diagram rewriting [Sel10; BSZ17]. These calculi have proved remarkably
 93 versatile, finding applications across a wide range of fields: *linear algebra* [BSZ17; Bon+19;
 94 Fre+25]; and in *optimisation and probability* [Ste+24; SS21; BDS21].

95 Throughout this work we explore how, step by step, one can enrich the already known theories
 96 for linear [BSZ17] algebra by adding one new generator at a time, along with a particular set of
 97 axioms, and derive even richer structures for convex problems [SS24].

98 The development followed a strict linear discipline, adding exactly one new generator and
 99 verifying that the resulting algebra is sound, expressive, and complete with respect to a precisely
 100 defined semantic category. Each new class of optimization problems then costs a single additional
 101 generator and a finite set of axioms. This is made precise by the normal form and completeness
 102 results of Theorem 3.3 and Theorem 4.1, which follow the same pattern already established
 103 for graphical linear and affine algebra [BSZ17; Bon+19] and for their quadratic [Ste+24] and
 104 polyhedral [BDS21] extensions. Table 1.1 summarizes the resulting hierarchy of languages,
 105 indicating where each syntax and axiomatisation is introduced.

Table 1.1 – Hierarchy of the graphical languages used in this thesis.

Language	Syntax	Axioms	Reference	Semantics
GLA	Appendix ??: linear generators	Appendix ??: linear equations	[BSZ17]	LinRel
GAA	GLA + $\vdash$; Table ??	GLA axioms + affine-point axioms; Appendix ??	[Bon+19]	AffRel
GPA	GAA + $\dashv\!\!\!\!\!\supseteq$; Definition 3.3	GAA axioms + order axioms; Definition 3.3	[BDS21]	PolyRel
GLP	GPA + $\sqsubset$; Definition 4.5	GPA axioms + objective axioms; Definition 4.5	This work	LinProg (LPRel)

106 We start by recalling how the Symmetric Monoidal Theory of Graphical Affine Algebra
 107 (**GAA**) [Bon+19] can be extended into the even more general theory of Symmetric Inequality
 108 Monoidal Theory. This is achieved by adding the inequality generator $\dashv\!\!\!\!\!\supseteq$. Geometrically, this
 109 allows our algebra to go beyond affine spaces and start representing polyhedral figures as linear
 110 inequality constraints in optimization.

$$\{|x \in P\} = \begin{cases} 0, & Ax + b \geq 0, \\ +\infty, & \text{otherwise.} \end{cases}$$

111 Inspired by these observations, and parallel works in this direction [Ste+24; Law73], we make
 112 our central contribution in the final chapter: the construction of a graphical algebra for parametric
 113 linear programming. In this setting, a linear program is represented as a morphism that can be
 114 composed with other optimization blocks, leading to the idea of *composing optimization problems*.
 115 This gives a symbolic language in which the elimination of internal variables corresponds directly
 116 to categorical composition, allowing optimization problems to be manipulated diagrammatically
 117 before computation.

118 For that we present the new prop of Parametric Linear Programmas where morphisms are
 119 linear programmas, this will serve as our semantics. Additionally, we make the addition of the new
 120 generator $\sqsubset$ and other three characteristic axioms, and through a epigraph-based argument,
 121 provide a normal form proof to show that this small set of generators and rules fully present the
 122 semantics category, i.e, it's sound, complete and expressible.

123 **Thesis overview** In Section 2.1 we set up the algebraic and categorical setting on which the rest
 124 of the work is built, introducing quantales and relations and the interchangeable role they play
 125 in a categorical view. Next, in Section 2.2, we apply this theory to convex optimization [SS24;
 126 Kad26], showing how several of its central constructs fall within the same categorical framework.

127 Chapter 3 presents the already-known graphical theory of Polyhedral Algebra [BDS21],
 128 where some of the known proofs are recalled and refined.

129 Finally, in Chapter 4, we present the central contribution of this thesis: an entirely new
 130 Graphical Algebra, **GLP**, dedicated to expressing parametric Linear Programs. Building on the
 131 polyhedral generators of Chapter 3, we introduce a single new generator for the linear objective
 132 function and derive a complete axiomatization for right-hand side parametric linear programming,
 133 proving that **GLP** presents the prop **LPRel**. This work represents a first step toward a complete
 134 diagrammatic calculus for convex optimization.

135 2 BACKGROUND

136 2.1 RELATIONS AND QUANTALES

137 In this section we explore the theory of relations. On set-theoretical foundations, relations are
 138 among the most primitive notions available, and yet they already support remarkably powerful
 139 abstractions: many other structures, functions among them, are derived from relations alone.
 140 Later we shall see that relations sit comfortably within the language of category theory, which in
 141 turn offers a natural vocabulary for describing them.

142 We begin with the most basic kind of relation:

143 **Definition 2.1** (Binary Relations). *Let X and Y be sets. A binary relation R from X to Y is a subset*

$$R \subseteq X \times Y.$$

144 *Equivalently, it can be identified with its characteristic function*

$$\chi_R : X \times Y \rightarrow \{0, +\infty\},$$

145 *defined by*

$$\chi_R(x, y) = \begin{cases} 0 & \text{if } (x, y) \in R, \\ +\infty & \text{otherwise.} \end{cases}$$

146 A binary relation specifies whether x is related to y . Observe that one can think of a relation
 147 in two different ways that will prove useful later. First, as a set of pairs, where every relation is
 148 just a pair $(x, y) \in R$, or as a map that has x, y as inputs and outputs 0 if they are related, and
 149 $+\infty$ if they are not the choice of ∞ is not arbitrary; normal choices here are actually $\{true, false\}$
 150 or $\{0, 1\}$ but ∞ already prepares the settings for the very next extensions discussed below.

151 This is a general construct as the sets X and Y can be anything. We can, for instance, ask for
 152 further structures on such sets and obtain different kinds of relations.

153 **Definition 2.2** (Linear/Affine Relations). *Let X and Y be vector spaces. A linear relation from X
 154 to Y is a linear subspace*

$$R \subseteq X \times Y.$$

155 *An affine relation is an affine subspace of $X \times Y$.*

156 *Equivalently, a linear relation can be described as the graph of a possibly multivalued linear*
 157 *operator, i.e.,*

$$R(x) := \{y \in Y \mid (x, y) \in R\}.$$

158 *or, its characteristic function defined by*

$$\chi_R : X \times Y \rightarrow \{0, +\infty\},$$

$$\chi_R(x, y) = \begin{cases} 0 & \text{if } (x, y) \in R, \\ +\infty & \text{otherwise.} \end{cases}$$

159 Thus, a linear/affine relation is a binary relation where the set R is a subspace

160 Notice, however, that we have been restricting the relations between x and y to be a case
161 of "they are, or they are not related". In fact, we might be interested in adding degrees to such
162 a relation, so that we could say that x and y are "more" or "less" related. This leads us to the
163 well-known notion of weighted relations [Law73].

164 **Definition 2.3** (Weighted Relations). *Let X and Y be any two sets and let $\overline{\mathbb{R}}^+ := [0, +\infty]$. A*
165 *weighted relation from X to Y is a set*

$$R \subseteq X \times Y \times \overline{\mathbb{R}}^+$$

166 *such that for every $(x, y) \in X \times Y$ there exists exactly one $r \in \overline{\mathbb{R}}^+$ satisfying*

$$(x, y, r) \in R.$$

167 *Equivalently, we may define the characteristic function*

$$\chi_R : X \times Y \rightarrow \overline{\mathbb{R}}^+$$

168 *by*

$$\chi_R(x, y) = r \quad \text{if and only if } (x, y, r) \in R.$$

169 and the codomain is not restricted to a binary set anymore. In fact, one could choose any other
170 set instead of the $\overline{\mathbb{R}}^+$ to be the codomain of the characteristic function as long as it satisfy some
171 properties. The whole structure of a relation is further generalized by the notion of a quantale.

172 **Definition 2.4** (Quantale). *A quantale is a structure $(Q, \leq, \otimes, e)$ such that:*

- 173 • *$(Q, \leq)$ is a complete lattice, i.e., every subset $A \subseteq Q$ admits a join $\bigvee A$ and a meet $\bigwedge A$;*
- 174 • *$(Q, \otimes, e)$ is a monoid;*
- 175 • *Multiplication distributes over arbitrary join in each variable:*

$$a \otimes \left(\bigvee_{i \in I} b_i \right) = \bigvee_{i \in I} (a \otimes b_i), \quad \left(\bigvee_{i \in I} a_i \right) \otimes b = \bigvee_{i \in I} (a_i \otimes b)$$

176 *for all families $\{a_i\}_{i \in I}, \{b_i\}_{i \in I} \subseteq Q$.*

177 *If $\otimes$ is commutative, the quantale is called commutative. If $e = \top$ (the top element of the*
178 *lattice), it is called integral.*

Each of these three axioms earns its place. The complete lattice gives meaning to the join $\vee$, the operation that eliminated the intermediate variable y in the transitivity example of the introduction. The monoid $(Q, \otimes, e)$ plays the role of combining evidence along a chain of relations, with e the trivial combination that changes nothing. Distributivity is what lets us freely interchange this join and this product, precisely the property that allows composition and variable elimination to be reduced to the same operation, rather than two operations that merely happen to look alike.

Given a quantale Q , we can define the notion of Q -valued relation [Coe+18].

Definition 2.5 (Generalized Relations). A generalized relation (or Q -valued relation) from X to Y , where Q is a quantale, is a set

$$R \subseteq X \times Y \times Q$$

such that for every $(x, y) \in X \times Y$ there exists exactly one $q \in Q$ satisfying

$$(x, y, q) \in R.$$

Its characteristic function is

$$\chi_R : X \times Y \rightarrow Q$$

defined by

$$\chi_R(x, y) = q \quad \text{if and only if } (x, y, q) \in R.$$

Now we are ready to build our previous relations as particular cases of this more general framework.

Example 2.1. The notion of a Q -valued relation recovers familiar concepts for particular choices of quantale.

(1) Truth quantale. Let $Q = \{0, 1\}$ with the usual order $0 \leq 1$, join given by $\vee$, and monoidal product $\otimes = \wedge$ with unit $e = 1$. This is the (Boolean) truth quantale.

A Q -valued relation is then a function

$$R : X \times Y \rightarrow \{0, 1\}.$$

Such a function is precisely the characteristic function of an ordinary binary relation

$$\tilde{R} \subseteq X \times Y, \quad (x, y) \in \tilde{R} \iff R(x, y) = 1.$$

Thus Q -valued relations coincide with classical binary relations.

This also settles the indicator convention adopted in Definition 2.1, where relations were valued in $\{0, +\infty\}$ rather than $\{0, 1\}$. The two are the same quantale: the map $1 \mapsto 0, 0 \mapsto +\infty$ sends $\vee$ to $\min$, $\wedge$ to $+$, and the unit 1 to the unit 0 , hence is an isomorphism of the truth quantale

204 onto the sub-quantale $\{0, +\infty\}$ of the Lawvere quantale below. We work with $\{0, +\infty\}$ throughout,
 205 so that Boolean relations sit inside the cost-valued ones without any change of convention.

206 **(2) Lawvere quantale.** Let $Q = \overline{\mathbb{R}}^+ = [0, +\infty]$ ordered by the reverse of the usual order (so that
 207 $a \leq b$ iff $a \geq b$ in the usual order), with monoidal product given by addition

$$a \otimes b := a + b,$$

208 and unit $e = 0$. Joins are then given by infima in the usual order.

209 This structure is the Lawvere quantale. A Q -valued relation

$$R : X \times Y \rightarrow \overline{\mathbb{R}}^+$$

210 is precisely a weighted relation assigning a (possibly infinite) cost to each pair (x, y) .

211 Hence generalized relations unify ordinary binary relations and weighted relations as special
 212 cases corresponding to particular quantales.

213 Note that linear and affine relations are not recovered by this construction, since they demand
 214 extra structure on the sets X and Y , specifically that R be a linear or affine subspace of $X \times Y$.
 215 They are, nonetheless, fully embedded in the framework of binary relations.

216 Next, we expand the current definition of Lawvere Quantales to the more general Extended-
 217 Lawvere Quantale:

218 **Definition 2.6** (Optimization/Extended Lawvere Quantale). *Let*

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$

219 The optimization quantale is the structure

$$(\overline{\mathbb{R}}, \geq, \otimes, e)$$

220 defined as follows:

- 221 • The order $\geq$ is the usual order on the extended real line. With this order, arbitrary joins
 222 are given by the usual infima:

$$\bigvee A = \inf A.$$

- 223 • The monoidal product is addition:

$$a \otimes b := a + b,$$

224 where addition is extended to $\overline{\mathbb{R}}$ by $a + (+\infty) = +\infty$ for every a , and $a + (-\infty) = -\infty$

225 for every $a \neq +\infty$. In particular we adopt the convention

$$(+\infty) + (-\infty) := +\infty,$$

226 so that $+\infty$ is absorbing. This is Moreau's inf-addition [Roc70], and it is the only extension
227 under which the structure below is a quantale.

228 • The unit element is

$$e := 0.$$

229 With these operations, $(\overline{\mathbb{R}}, \geq, +, 0)$ is a commutative quantale. Joins are ordinary infima, and
230 addition distributes over arbitrary joins.

231 The convention above is not a matter of taste. Without it, $+$ is only partially defined on $\overline{\mathbb{R}}$, so
232 $(\overline{\mathbb{R}}, +, 0)$ fails to be a monoid and the quantale axioms cannot even be stated.

233 **Proposition 2.1.** $(\overline{\mathbb{R}}, \geq, +, 0)$ with inf-addition is a commutative quantale.

234 *Proof.* Since $\overline{\mathbb{R}}$ under the reversed order is a complete lattice with $\bigvee A = \inf A$, $\perp = \bigvee \emptyset = +\infty$
235 and $\top = -\infty$, only the monoid structure and distributivity need checking.

236 **(1) Monoid.** Inf-addition is commutative by construction and total, since the one ambiguous
237 sum is now assigned the value $+\infty$. For associativity, note that $a + b = +\infty$ exactly when
238 $a = +\infty$ or $b = +\infty$, and $a + b = -\infty$ exactly when neither is $+\infty$ and at least one is $-\infty$. Both
239 conditions are symmetric in the summands and depend only on the multiset $\{a, b\}$, so $(a + b) + c$
240 and $a + (b + c)$ agree on every triple: each equals $+\infty$ if some argument is $+\infty$, else $-\infty$ if
241 some argument is $-\infty$, else the ordinary finite sum. The unit is 0.

242 **(2) Distributivity.** Let $a \in \overline{\mathbb{R}}$ and $\{b_i\}_{i \in I}$ be any family. We must show $a + \inf_i b_i = \inf_i (a + b_i)$,
243 where both sides are $+\infty$ when $I = \emptyset$.

244 If $a = +\infty$, both sides are $+\infty$ by absorption. If a is finite, the identity is the standard
245 translation invariance of the infimum. If $a = -\infty$ and every $b_i = +\infty$, then $\inf_i b_i = +\infty$ and
246 both sides are $+\infty$; if instead some $b_{i_0} < +\infty$, then $\inf_i b_i < +\infty$, so the left side is $-\infty$, and
247 the right side is $-\infty$ as well because $a + b_{i_0} = -\infty$. Commutativity gives distributivity in the
248 other variable. $\square$

249 **Definition 2.7** (Optimization (Cost/Reward) Relations). Let X and Y be sets. An optimization
250 relation from X to Y is an $\overline{\mathbb{R}}$ -valued relation, i.e., a function

$$R : X \times Y \rightarrow \overline{\mathbb{R}}.$$

251 Equivalently, it is a subset

$$R \subseteq X \times Y \times \overline{\mathbb{R}}$$

252 such that for every $(x, y) \in X \times Y$ there exists exactly one $r \in \overline{\mathbb{R}}$ satisfying

$$(x, y, r) \in R.$$

253 The value $R(x, y)$ is interpreted as the objective value (cost or reward) associated to the pair
254 (x, y) , with $+\infty$ and $-\infty$ representing extremal infeasibility or unboundedness.

255 Observe how this is essentially the weighted relations extended along the negative values.
256 The importance of it is that we want to be able to express optimization problems that might be
257 unfeasible ($+\infty$) or unbounded ($-\infty$), so we need to add both the top and bottom element.

258 All of this admits a suitable and natural categorical construction. Specifically, from any
259 quantale Q one can not only define Q -relations, but also construct an entire category of them.
260 This category is the natural setting for discussing the central concept of this framework: the
261 composition of relations.

262 The following lemma shows how to build such a category and establishes that this category
263 also has a very important monoidal structure behind it. Particularly, the category is a compact
264 closed symmetric monoidal category (for the definition of such see ??) [Lan98]:

265 **Proposition 2.2** (Category $\mathbf{Rel}(Q)$). *For a quantale Q , the Q -relations form a category $\mathbf{Rel}(Q)$
266 with composition and identities*

$$(S \circ R)(a, c) = \bigvee_b R(a, b) \otimes S(b, c) \quad 1_A(a, b) = \bigvee \{k \mid a = b\}$$

267 If Q is a commutative quantale, then $\mathbf{Rel}(Q)$ carries a symmetric monoidal structure, with
268 the categorical monoidal product $\otimes_{\mathcal{C}}$ on objects given by the Cartesian product of sets and the
269 action on relations given for $R : A \rightarrow C$ and $S : B \rightarrow D$ by

$$(R \otimes_{\mathcal{C}} S)(a, b, c, d) = R(a, c) \otimes S(b, d)$$

270 The singleton set is the monoidal unit, and the category is compact closed w.r.t the monoidal
271 structure

272 *Proof.* We verify the categorical structure.

273 **(1) Well-defined composition.** Let $R : A \rightarrow B$ and $S : B \rightarrow C$ be Q -relations. For each
274 $(a, c) \in A \times C$, the set

$$\{R(a, b) \otimes S(b, c) \mid b \in B\} \subseteq Q$$

275 admits a supremum since Q is a complete lattice. Hence $(S \circ R)(a, c)$ is well-defined as an element
276 of Q .

277 **(2) Associativity.** Let $R : A \rightarrow B$, $S : B \rightarrow C$, and $T : C \rightarrow D$. For $a \in A$, $d \in D$,

$$((T \circ S) \circ R)(a, d) = \bigvee_c \left(\bigvee_b R(a, b) \otimes S(b, c) \right) \otimes T(c, d).$$

278 Since $\otimes$ distributes over arbitrary suprema,

$$= \bigvee_{b,c} R(a, b) \otimes S(b, c) \otimes T(c, d).$$

279 Similarly,

$$(T \circ (S \circ R))(a, d) = \bigvee_b R(a, b) \otimes \left(\bigvee_c S(b, c) \otimes T(c, d) \right) = \bigvee_{b,c} R(a, b) \otimes S(b, c) \otimes T(c, d).$$

280 Thus composition is associative.

281 **(3) Identities.** For each set A , define

$$1_A(a, b) = \begin{cases} e & \text{if } a = b, \\ \perp & \text{otherwise,} \end{cases}$$

282 where $\perp = \bigvee \emptyset$ is the bottom element of Q .

283 Then for $R : A \rightarrow B$,

$$(R \circ 1_A)(a, b) = \bigvee_{a'} 1_A(a, a') \otimes R(a', b).$$

284 All terms vanish except when $a' = a$, so

$$= e \otimes R(a, b) = R(a, b).$$

285 Similarly, $1_B \circ R = R$. Hence identities exist.

286 **(4) Monoidal structure (commutative case).** Assume Q is commutative. Define the tensor on
287 objects by Cartesian product:

$$A \otimes_{\mathcal{C}} B := A \times B.$$

288 For relations $R : A \rightarrow C$ and $S : B \rightarrow D$, define

$$(R \otimes_{\mathcal{C}} S)((a, b), (c, d)) = R(a, c) \otimes S(b, d).$$

289 Functoriality follows from associativity and commutativity of $\otimes$:

$$(S_1 \otimes S_2) \circ (R_1 \otimes R_2) = (S_1 \circ R_1) \otimes (S_2 \circ R_2).$$

290 The singleton set 1 is a unit since $A \times 1 \cong A$.

291 Symmetry is induced by the swap map

$$\sigma_{A,B}((a, b), (b', a')) = \begin{cases} e & \text{if } a = a', b = b', \\ \perp & \text{otherwise.} \end{cases}$$

292 **(5) Compact closure.** For each set A , define unit and counit relations

$$\eta : 1 \rightarrow A \times A, \quad \epsilon : A \times A \rightarrow 1$$

293 by

$$\eta(*, (a, a')) = \begin{cases} e & a = a', \\ \perp & \text{otherwise,} \end{cases} \quad \epsilon((a, a'), *) = \begin{cases} e & a = a', \\ \perp & \text{otherwise.} \end{cases}$$

294 A direct computation using the definition of composition shows that the triangle identities hold:

$$(1_A \otimes \epsilon) \circ (\eta \otimes 1_A) = 1_A, \quad (\epsilon \otimes 1_A) \circ (1_A \otimes \eta) = 1_A.$$

295 Thus every object is self-dual, and $\mathbf{Rel}(Q)$ is compact closed. □

296 A curious fact about relations is that they actually encode the notion of matrices and matrix
297 multiplication. This serves as motivation for what will later become the diagrammatical theory
298 of linear algebra. The following remark expands on it showing that every morphism in $\mathbf{Rel}(Q)$
299 corresponds to a matrix with entries in Q , and that composition is exactly matrix multiplication.

300 **Remark 2.1.** A Q -valued relation $R : X \times Y \rightarrow Q$ is nothing but a Q -valued matrix indexed by
301 $X \times Y$: one simply reads off the entry at position (x, y) as $\chi_R(x, y) \in Q$. That is, a generalized
302 relation is precisely an element of $Q^{X \times Y}$.

303 When X and Y are finite sets, say $|X| = m$ and $|Y| = n$, this becomes completely explicit: a
304 Q -valued relation is exactly an $m \times n$ matrix

$$M_R = [\chi_R(x_i, y_j)]_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}} \in Q^{m \times n},$$

305 whose (i, j) -entry records the Q -value assigned to the pair (x_i, y_j) .

306 Moreover, composition of Q -valued relations corresponds precisely to matrix multiplication in
307 Q . Given a second Q -valued relation $S : Y \times Z \rightarrow Q$, their composite $S \circ R : X \times Z \rightarrow Q$ is
308 defined by

$$\chi_{S \circ R}(x, z) = \bigvee_{y \in Y} \chi_R(x, y) \otimes \chi_S(y, z),$$

309 which, in matrix notation, reads

$$M_{S \circ R} = M_R \otimes M_S, \quad (M_R \otimes M_S)_{ij} = \bigvee_{k=1}^n (M_R)_{ik} \otimes (M_S)_{kj}.$$

310 *This is exactly the usual row-times-column matrix product, but with ordinary addition replaced*
 311 *by the join $\vee$ and ordinary multiplication replaced by the monoidal product $\otimes$ of Q . The two*
 312 *axioms of a quantale — that $\otimes$ distributes over arbitrary joins — are precisely what guarantees*
 313 *that this multiplication is associative and well-defined, so that Q -valued matrices form a category*
 314 *under this composition.*

315 To illustrate these new concepts, we look at some of our previous relations now built as
 316 categories.

317 **Example 2.2** (Category of Boolean relations). *Let*

$$Q = (\{0, 1\}, \leq, \otimes, e)$$

318 *be the Boolean quantale where:*

- 319 • $0 \leq 1$;
- 320 • $\otimes = \wedge$ (logical conjunction);
- 321 • $e = 1$;
- 322 • joins are given by $\vee$.

323 *A Q -valued relation $R : A \rightarrow B$ is a function*

$$R : A \times B \rightarrow \{0, 1\},$$

324 *which is precisely the characteristic function of a subset*

$$\tilde{R} \subseteq A \times B.$$

325 *The composition formula in $\mathbf{Rel}(Q)$ becomes*

$$(S \circ R)(a, c) = \bigvee_b R(a, b) \wedge S(b, c).$$

326 *Since joins are logical disjunction, this reduces to*

$$(S \circ R)(a, c) = 1 \iff \exists b \in B \text{ such that } R(a, b) = 1 \text{ and } S(b, c) = 1.$$

327 *Thus composition coincides exactly with the usual composition of binary relations.*

328 *Since the Boolean quantale is commutative, $\mathbf{Rel}(Q)$ carries a symmetric monoidal structure:*

- 329 • *On objects:*

$$A \otimes B := A \times B.$$

- On morphisms:

$$(R \otimes S)((a, b), (c, d)) = R(a, c) \wedge S(b, d).$$

The singleton set is the monoidal unit. With this structure, $\mathbf{Rel}(Q)$ is precisely the classical symmetric monoidal compact closed category $\mathbf{Rel}$.

Example 2.3 (Category of weighted relations). Let

$$Q = (\mathbb{R} \cup \{+\infty\}, \leq_Q, \otimes, e)$$

be the Lawvere quantale where:

- $a \leq_Q b \iff a \geq b$ in the usual order;
- $\otimes = +$;
- $e = 0$;
- joins are given by infima in the usual order.

A Q -valued relation $R : A \rightarrow B$ is a function

$$R : A \times B \rightarrow \mathbb{R} \cup \{+\infty\},$$

assigning a cost (or weight) to each pair (a, b) .

The composition formula becomes

$$(S \circ R)(a, c) = \inf_{b \in B} (R(a, b) + S(b, c)).$$

Thus composition is given by infimal convolution [SS24].

Since addition is commutative, the Lawvere quantale is commutative, and $\mathbf{Rel}(Q)$ carries a symmetric monoidal structure:

- On objects:

$$A \otimes B := A \times B.$$

- On morphisms:

$$(R \otimes S)((a, b), (c, d)) = R(a, c) + S(b, d).$$

The singleton set is the monoidal unit. With this tensor product, $\mathbf{Rel}(Q)$ is a symmetric monoidal compact closed category of weighted relations.

In particular:

- If A, B, C are finite sets, relations may be viewed as matrices with entries in $\mathbb{R} \cup \{+\infty\}$.

- *Composition is matrix multiplication in the $(\min, +)$ -semiring.*

Hence $\mathbf{Rel}(Q)$ is the category of weighted relations, which underlies dynamic programming, tropical algebra, and shortest-path problems.

Those examples illustrate how naturally the quantale structure supports categorical constructions: composition of Q -valued relations corresponds to matrix multiplication driven by $\otimes$, and the tensor product of relations is governed by the join $\vee$, both operations coming directly from the quantale axioms.

The resulting categories are symmetric monoidal, and it is precisely this monoidal backbone that will support the algebraic structure we aim to build. A key piece of that structure is a *Frobenius monoid* on each object: a monoid (μ, η) and a comonoid (δ, ϵ) on the same object, satisfying the Frobenius law

$$(\mathrm{id} \otimes \mu) \circ (\delta \otimes \mathrm{id}) = \delta \circ \mu = (\mu \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \delta).$$

A symmetric monoidal category in which every object carries such a structure, compatibly with the tensor product, is called a *hypergraph category* [FS19].

Hypergraph categories are related to, but strictly more general than, compact closed categories [Sel10]. In a compact closed category every object A has a dual A^* with unit $\eta_A : I \rightarrow A^* \otimes A$ and counit $\epsilon_A : A \otimes A^* \rightarrow I$ satisfying the snake equations; a Frobenius structure does not require the existence of such a dual. Conversely, if a compact closed category is such that every object is self-dual ($A \cong A^*$) and the induced cups and caps satisfy the Frobenius law, then it is a hypergraph category. Thus compact closed categories with self-dual objects form a special class of hypergraph categories, and it is the later, more general, notion that we shall adopt as our ambient framework.

Remark 2.2. *If Q is commutative, then $\mathbf{Rel}(Q)$ is a hypergraph category and admits a commutative Frobenius structure on every object. Diagrammatically, we denote the additive and co-additive structure as*

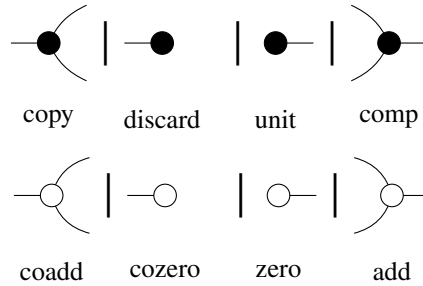
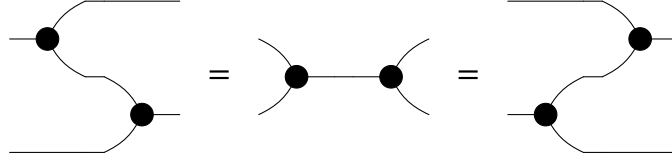


Figure 2.1 – Frobenius additive(black) and co-additive(white) structure

Subject to the Frobenius Law (both colors)



376 *Proof.* Just a check that all laws do hold for any $\mathbf{Rel}(Q)$ category □

377 These constructions provide exactly the semantic foundation for the algebra we aim to build.
 378 In the same way that semantics is understood in linguistics and programming language theory
 379 [Win93], the semantics of a formal system assigns meaning to its expressions. More precisely,
 380 while syntax consists of the tokens and rules for forming and combining expressions, semantics
 381 is the interpretation assigned to each such expression — in our case, an optimization problem
 382 living in $\mathbf{Rel}(Q)$.

383 As outlined in the introduction, the central goal of this work is to show that optimization
 384 emerges naturally from algebraic and categorical structure. The key mechanism enabling this
 385 is the infimal composition present in the definition of the Costs/Rewards category: composing
 386 two relations automatically performs variable elimination, which is precisely the fundamental
 387 operation underlying optimization.

388 To make this precise, we need both a semantics — provided by the categorical framework
 389 developed above — and a syntax: a rigorous, compositional language for expressing optimization
 390 problems symbolically. The map between those is structure-preserving map, a functor, from the
 391 syntactic layer into the semantic one.

392 The syntactic layer will be formalized as a Symmetric Monoidal Theory (SMT) [BSZ17],
 393 which is a presentation of a category by generators and equations, whose terms respect the
 394 compositional structure of a symmetric monoidal category. This is the standard tool for building
 395 graphical languages in the style of Graphical Linear Algebra, and it is the framework within
 396 which our completeness results will be stated and proved.

397 **Definition 2.8** (Symmetric Monoidal Theory). *A SMT is a pair (Σ, E) , such that, Σ is a signature*
 398 *of operators o that has arity m and coarity n*

399 *Σ -terms are defined inductively as the smallest collection of typed expressions $t : m \rightarrow n$*
 400 *such that:*

- 401 1. (Generators) *If $o \in \Sigma$ with $o : m \rightarrow n$, then o is a Σ -term of type $m \rightarrow n$.*
- 402 2. (Identity) *For every $n \in \mathbb{N}$, there is a term*

$$\text{id}_n : n \rightarrow n.$$

- 403 3. (Composition) *If*

$$a : m \rightarrow n \quad \text{and} \quad b : n \rightarrow k$$

404 are Σ -terms, then

$$a; b : m \rightarrow k$$

405 is a Σ -term.

406 4. (Tensor product) If

$$a : m_1 \rightarrow n_1 \quad \text{and} \quad b : m_2 \rightarrow n_2$$

407 are Σ -terms, then

$$a \otimes b : m_1 + m_2 \rightarrow n_1 + n_2$$

408 is a Σ -term.

409 5. (Symmetry) For all $m, n \in \mathbb{N}$, there is a term

$$\sigma_{m,n} : m + n \rightarrow n + m.$$

410 and E be a set of equations over the Σ – terms.

411 Often, we will think of elements $o : m \rightarrow n \in \Sigma$ as generators and represent them as
 412 string diagrams. Being the Σ – terms the results of composing those generators horizontally and
 413 vertically inductively. The set E as a collection of axiomatic equations over these string diagrams.

414 The categorical counterpart of these terms is the free category generated by the SMT.

415 **Definition 2.9** (The Free category of SMT). *The $\text{FreeP}_{(\Sigma, E)}$ is the category freely generated by*
 416 *(Σ, E) where the objects are natural numbers, morphisms are Σ – terms quotiented by E , with*
 417 *composition and tensor product inherited from the symmetric monoidal structure. $\text{FreeP}_{(\Sigma, E)}$ is*
 418 *in fact a PROP¹.*

419 The $\text{FreeP}_{(\Sigma, E)}$, also denoted as $P\Sigma_{/E}$,² will be our **syntax** responsible for expressing our
 420 underlying semantics. For it to be considered a "useful" syntax we expect that it satisfies some
 421 properties, particularly we want it to be complete, sound and expressive.

422 Without those, our algebra would be empty since we would have no assurances that it actually
 423 says anything meaningful about we desire to say.

424 **Definition 2.10.** *We will say that a SMT axiomatizes or represents a specific category, usually*
 425 *called the semantic category, when the following diagram commutes:*

¹For the definition of a prop see [Lan98]

²During the work we will use FreeP_S and S , for whatever syntax S may be, interchangeably.

$$\begin{array}{ccc}
P\Sigma_{/E} = \text{Free}P_{(\Sigma, E)} & \xrightarrow{\cong} & \text{Im}(\llbracket \cdot \rrbracket) \\
\uparrow q & \nearrow s & \downarrow i \\
\text{Syn} = P\Sigma & \xrightarrow{\llbracket \cdot \rrbracket} & \text{Sem}
\end{array}$$

where: $P\Sigma$ is the category where objects are natural numbers and morphisms are Σ – terms from $n \rightarrow m$; $P\Sigma_{/E} = \text{Free}P_{(\Sigma, E)}$ is the $P\Sigma$ category with the morphisms quotiented by E and there is then a prop morphism $q : P\Sigma \rightarrow P\Sigma_{/E}$ witnessing this quotient; **Sem** is our semantic category (the Cost/Reward relations for example); and s is a full and faithful prop morphism. Another way to say the same is to demand two things out of the SMT, namely:

- *Soundness*: We say that $P\Sigma_{/E}$ is sound if $c \stackrel{E}{=} d^3$ implies $[c] = [d]$.
- *Completeness*: We say that $P\Sigma_{/E}$ is complete if $[c] = [d]$ implies $c \stackrel{E}{=} d$
- *Expressivity*: We say that $P\Sigma_{/E}$ is expressive if for every morphism c in **Sem** there exists $d \in P\Sigma_{/E}$ such that $[d] = c$

Those three properties capture the idea that everything we can prove in the syntax, is a holds in the semantics (soundness), everything that holds in the semantics is provable in the syntax (completeness) and that every morphism in the semantics has a syntactic representative (expressivity). Moreover, these properties denote equivalent characteristics about the interpretation map, both categorically and functionally. Below we depict a small table defining the equivalence between the desired properties and their equivalent statement:

Now we introduce two of the foundational theories for Graphical Algebra, that will serve as a foundation for the following sections, namely, Graphical Linear Algebra [PRS22; BSZ17] and Graphical Affine Algebra [Bon+19], we omit their definitions and proofs and refer to the original sources.

However, we do provide some notation on how to talk about those during the course of the work. Notice that, their semantics was already introduced when we talked about Linear/Affine relations.

Definition 2.11. We will call **GLA** the presentation of the Graphical Linear Algebra SMT, and, in the same manner, **GAA** the SMT of Graphical Affine Algebra.

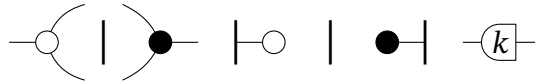
Definition 2.12. We also denote with $\overrightarrow{(\cdot)}/\overleftarrow{(\cdot)}$ when we want to talk about the algebra generated without the Frobenius structure. For example $\overrightarrow{\text{GLA}}$ is the algebra generated by

³ $\stackrel{E}{=}$ means that this equality is provable by the equations in E

Table 2.1 – Equivalence of properties for the interpretation map $\llbracket - \rrbracket : P\Sigma/E \rightarrow \mathbf{Sem}$

Set theory	Category theory	Logic (SMT)
Total	Functor	Soundness: $c \stackrel{E}{=} d \Rightarrow \llbracket c \rrbracket = \llbracket d \rrbracket$
Deterministic	Well-defined map	(Functionality of the interpretation)
Injective	Faithful	Completeness: $\llbracket c \rrbracket = \llbracket d \rrbracket \Rightarrow c \stackrel{E}{=} d$
Surjective	Full	Expressivity: $\forall f \in \mathbf{Sem}, \exists d \in P\Sigma/E \text{ s.t. } \llbracket d \rrbracket = f$

452 *without their symmetric*



453 *and the exactly opposite we denote as $\overleftarrow{\mathbf{GLA}}$.*

454 2.2 COMPOSABILITY OF CONVEX PROBLEMS

455 Having established the necessary categorical and algebraic tools, we proceed to their applica-
 456 tion in convex analysis. In this section we introduce an initial categorical framework for convex
 457 optimization [SS24], showing how the fundamental operations of convex analysis — infimal
 458 convolution, conjugation, and variable elimination — arise naturally as categorical constructions.
 459 This gives the first concrete evidence that optimization is not merely a collection of isolated
 460 techniques, but a coherent algebraic theory with deep compositional structure.

461 The progression of the section is as follows. We begin with the general category of bifunctions,
 462 which serves as the broadest setting, and progressively restrict to convex bifunctions, where
 463 the analytical structure is richest. These results lay the groundwork for the graphical theories
 464 developed in the following sections: Polyhedral Algebra and Graphical Linear Programming will
 465 both arise as full subcategories of $\mathbf{CxBiFn}$, each corresponding to a class of bifunctions with
 466 additional algebraic structure.

467 Later, we will see the particular applications of such definitions and results. For now, however,
 468 we begin by defining the broadest category in the hierarchy, which will generalize most of the
 469 later work: the category of bifunctions.

470 **Definition 2.13** (Bifunction Category). *The category $\mathbf{BiFunc}$ is defined as follows:*

471 • *Objects are sets.*

472 • *A morphism $F : A \rightarrow B$ is a function*

$$F : A \times B \rightarrow \overline{\mathbb{R}}.$$

473 • *Given $F : A \rightarrow B$ and $G : B \rightarrow C$, their composition $G \circ F : A \rightarrow C$ is defined pointwise*
 474 *by*

$$(G \circ F)(a, c) := \inf_{b \in B} (F(a, b) + G(b, c)).$$

475 • *The identity morphism on an object A is the function*

$$1_A : A \times A \rightarrow \overline{\mathbb{R}}$$

476 *given by*

$$1_A(a, a') = \begin{cases} 0 & \text{if } a = a', \\ +\infty & \text{otherwise.} \end{cases}$$

477 *If we equip **BiFunc** with a monoidal structure:*

478 • *The tensor product on objects is the Cartesian product*

$$A \otimes B := A \times B.$$

479 • *For morphisms $F : A \rightarrow C$ and $G : B \rightarrow D$, their tensor product*

$$F \otimes G : A \times B \rightarrow C \times D$$

480 *is defined by*

$$(F \otimes G)((a, b), (c, d)) := F(a, c) + G(b, d).$$

481 *The singleton set serves as the monoidal unit.*

482 **Remark 2.3.** *The category **BiFunc** is isomorphic to the category **RelQ** for a suitable choice of*
 483 *Q .*

484 *Consider the quantale*

$$Q = (\overline{\mathbb{R}}, \leq_Q, \otimes, e)$$

485 *where:*

486 • $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\};$

- 487 • the order is reversed with respect to the usual order,

$$a \leq_Q b \iff a \geq b \quad (\text{usual order});$$

- 488 • the monoidal product is addition,

$$a \otimes b := a + b;$$

- 489 • the unit is $e = 0$.

490 In this quantale, arbitrary joins are given by infima in the usual order:

$$\bigvee_Q A = \inf A.$$

491 Now a Q -valued relation $R : A \rightarrow B$ is precisely a function

$$R : A \times B \rightarrow \overline{\mathbb{R}},$$

492 i.e. a bifunction. Moreover, the composition formula in $\mathbf{Rel}(Q)$ becomes

$$(S \circ R)(a, c) = \bigvee_b R(a, b) \otimes S(b, c) = \inf_b (R(a, b) + S(b, c)),$$

493 which is exactly the composition law in $\mathbf{BiFunc}$.

494 Therefore, $\mathbf{BiFunc}$ is (strictly) the same category as $\mathbf{Rel}(Q)$ for the Lawvere-type quantale
495 $(\overline{\mathbb{R}}, \geq, +, 0)$.

496 Bifunctions are fundamental objects in convex analysis [Roc70]. They provide a unified
497 language for expressing objective functions, feasibility constraints, and duality simultaneously.
498 The following proposition confirms that the framework is conservative: classical set-theoretic
499 relations embed faithfully into this richer setting.

500 **Proposition 2.3.** *The category of Binary Relations, i.e. the Boolean-relation, is embedded as a*
501 *full subcategory of $\mathbf{BiFunc}$.*

502 *Proof.* Just consider the functor $F : \mathbf{Rel}(B) \rightarrow \mathbf{BiFunc}$ that takes every morphism $R : X \rightarrow Y$,
503 which is a subset of $X \times Y$, and maps it into the characteristic function $\chi_R : X \times Y \rightarrow \{0, +\infty\}$,
504 also known as the indicator function $1_R(x, y)$, or in bracket notation $\{|(x, y) \in R|\}$ $\square$

505 Having established the general framework, we now restrict our attention to the convex setting,
506 which is the natural home for optimization. The key insight is that convexity is preserved under
507 infimal composition, so the convex bifunctions form a subcategory of $\mathbf{BiFunc}$ that is closed
508 under the categorical operations.

509 **Definition 2.14** (Convex Bifunction Category $\mathbf{CxBiFn}$). *Let*

$$Q := (\overline{\mathbb{R}}, \leq_Q, \otimes, e)$$

510 *be the extended-Lawvere quantale where:*

- 511 • $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\};$
- 512 • $a \leq_Q b \iff a \geq b$ in the usual order;
- 513 • $a \otimes b := a + b;$
- 514 • $e := 0.$

515 *The category $\mathbf{CxBiFn}$ is defined as follows:*

- 516 • **Objects:** finite-dimensional real vector spaces.
- 517 • **Morphisms:** A morphism $F : A \rightarrow B$ is a function

$$F : A \times B \rightarrow \overline{\mathbb{R}}$$

518 *satisfying:*

- 519 – F is proper (not identically $+\infty$);
- 520 – F is convex as a function on the product space $A \times B$;
- 521 – F is lower semicontinuous.
- 522 • **Composition:** Given $F : A \rightarrow B$ and $G : B \rightarrow C$, their composite

$$G \circ F : A \rightarrow C$$

523 *is defined by infimal convolution:*

$$(G \circ F)(a, c) := \inf_{b \in B} (F(a, b) + G(b, c)).$$

- 524 • **Identity:** For each object A , the identity morphism

$$1_A : A \rightarrow A$$

525 *is given by*

$$1_A(a, a') = \begin{cases} 0 & \text{if } a = a', \\ +\infty & \text{otherwise.} \end{cases}$$

• *Monoidal product*

– *On objects:*

$$A \otimes B := A \times B.$$

– *On morphisms:* For $F : A \rightarrow C$ and $G : B \rightarrow D$,

$$(F \otimes G)((a, b), (c, d)) := F(a, c) + G(b, d).$$

– *The monoidal unit is the zero-dimensional vector space $\{0\}$.*

Remark 2.4. *One may define the analogous category $\mathbf{CvBiFn}$ which is the category of concave bifunctions. For doing so, one should only change the composition from $\inf$ to $\sup$, this is:*

$$(G \circ F)(a, c) := \sup_{b \in B} (F(a, b) + G(b, c)).$$

Every result for $\mathbf{CxBiFn}$ will hold, in an equivalent way, to the category $\mathbf{CvBiFn}$

Remark 2.5. *Every finite-dimensional real vector space V is linearly isomorphic to $\mathbb{R}^n$, where $n = \dim V$.*

Hence, one may restrict the objects of $\mathbf{CxBiFn}$ to the family

$$\text{Ob}(\mathbf{CxBiFn}) = \{\mathbb{R}^n \mid n \in \mathbb{N}\}.$$

With this choice, $\mathbf{CxBiFn}$ becomes a skeletal category, since $\mathbb{R}^n \cong \mathbb{R}^m$ implies $n = m$.

Also, if one considers the isomorphism $n \leftrightarrow \mathbb{R}^n$, $\mathbf{CxBiFn}$ becomes a prop, this can be made strict by defining the monoidal product on objects as sum.

$$m \otimes n = m + n$$

which is isomorphic to

$$\mathbb{R}^m \otimes \mathbb{R}^m = \mathbb{R}^m \times \mathbb{R}^m = \mathbb{R}^{m+n}$$

Observe that we have constructed a category whose morphisms are interpreted as convex optimization problems. Since every convex bifunction $F : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ can be written as

$$F(x, y) = \inf_z z \quad \text{s.t. } z = F(x, y)$$

This observation is relevant since it shows that there's some sort of duality between relations and optimization, where a convex relation R is a bifunction of the form above. The compositional structure yields even more substantive insights.

$$(S \circ R)(x, z) = \inf_y \{R(x, y) + S(y, z)\}$$

545 Which shows that relational composition is precisely variable elimination: composing two
 546 relations automatically marginalises over the intermediate variable y by minimising the combined
 547 cost. In the binary case this reduces to the existential question of whether a mediating element
 548 exists; in the weighted case it asks for the *optimal* such element. Optimization is therefore not
 549 imposed on the categorical structure from outside — it emerges from composition itself.

550 Convex theory also has a very nice duality theory which is of particular interest in algebraic
 551 constructs such as this one. We, however, do not include such results in this thesis, leaving it for
 552 the interested reader [SS24; Wil15]

553 The results of this section establish **CxBiFn** as a well-behaved categorical setting for convex
 554 optimization. The special classes of morphisms of polyhedral bifunctions inherit this structure
 555 while admitting a finite, explicit representation. It is precisely this finiteness that makes a
 556 graphical algebra possible: morphisms can be encoded as diagrams composed from a finite set
 557 of generators, and the equational theory of the diagrams corresponds exactly to the equivalences
 558 between optimization problems. The following section make this precise.

559 3 POLYHEDRAL RELATIONS

560 We now turn to a combinatorially richer class of objects: polyhedra and polyhedral cones.
 561 These are the fundamental geometric structures underlying linear programming, and they admit a
 562 categorical treatment entirely analogous to the one developed for linear and affine relations. The
 563 key difference is that composition here is governed by polyhedron projection onto a subspace,
 564 which is precisely variable elimination, and it is this operation that categorical composition
 565 will simulate diagrammatically. One particular way to perform polyhedral projection is the
 566 Fourier-Motzkin Elimination (FME), which is the algorithm explored in this section to show,
 567 diagrammatically, that the composition of polyhedrals yields another polyhedral.

568 The theory developed here remains within the category of binary relations, since morphisms
 569 are still indicator functions of subsets of $\mathbb{R}^n \times \mathbb{R}^m$, encoding feasibility constraints rather than
 570 objective functions. In this sense, **GPA** does not yet enter the weighted relational world, and so
 571 is not yet able to express more meaningful optimization problems. This is nonetheless a critical
 572 step in the development. Axiomatizing linear inequality constraints diagrammatically, **GPA**
 573 provides the Symmetric Monoidal Inequality Theory (SMIT) [BDS21] that will serve as the
 574 structural foundation for our final contribution, **GLP**, where these constraints are combined with
 575 a linear objective to form full parametric linear programs.

576 In what follows we describe and show the complete theory of polyhedral relations and
 577 polyhedral cones. Just as **AffRel** captured affine space as morphisms, the categories **PCRel**
 578 and **PolyRel** capture polyhedral cones and polyhedra as morphisms, with relational composition
 579 playing the role of variable elimination.

580 3.1 SEMANTICS: THE PROP PCREL OF POLYHEDRAL CONES

581 We begin by recalling the standard definitions of polyhedral cones and the corresponding
 582 semantic category.

583 **Definition 3.1.** A set $C \subseteq \mathbb{R}^n$ is called a polyhedral cone if there exists a matrix $A \in \mathbb{R}^{m \times n}$ such
 584 that

$$C = \{x \in \mathbb{R}^n \mid Ax \geq 0\}.$$

585 **Definition 3.2** (The prop **PCRel** of polyhedral cones). Let k be an ordered field, in this case
 586 the real numbers. The prop **PCRel** _{k} is defined as follows:

- 587 • Objects are natural numbers $n \in \mathbb{N}$, interpreted as the space k^n .
- 588 • A morphism $n \rightarrow m$ is a polyhedral cone $R \subseteq k^n \times k^m$ for which there exist $p \in \mathbb{N}$ and

589 $A \in k^{p \times (n+m)}$ such that

$$R = \left\{ (x, y) \in k^n \times k^m \mid A \begin{pmatrix} x \\ y \end{pmatrix} \geq 0 \right\}.$$

- 590 • *Composition is relational composition.*
- 591 • *The monoidal product on objects is addition $n \otimes m := n + m$, and on morphisms is the*
- 592 *Cartesian product.*

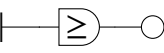
593 Observe that in comparison with the previous categories of **AffRel**, this category is way more
 594 general although it does not completely encode the notion of Affine Spaces. However, it does
 595 introduce to us a somewhat unorthodox notion of conical polyhedral relations. That is, the fact
 596 that two elements x, y may be related to one another by a relation R such that the set of all pairs
 597 xRy form a polyhedral cone. Although this description remains very set-theoretically, it starts
 598 to flirt with the idea of x and y being apparted from a specific class of maps. More precisely,
 599 instead of thinking of the whole geometry of polyhedral cones, one may think of the pairs (x, y)
 600 such that $Ax \geq By$ for some A and B fully defined by the relation R .

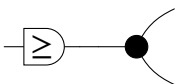
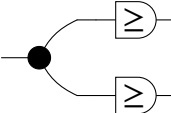
601 3.2 SYNTAX: THE ALGEBRA $\mathbf{GPA}_{\perp}$

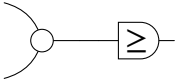
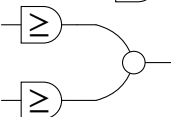
602 **Definition 3.3** (Graphical Polyhedral Algebra). *We call **GPA** the SMT generated by extending*
 603 ***GAA** with the following generator and its semantics:*

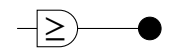
$$\llbracket \text{---} \boxed{\geq} \text{---} \rrbracket = \{(x, y) \mid x, y \in k, x \geq y\}$$

604 *along with the following additional equations:*

(P0)  = 

(P1)  $\subseteq$ 

(P2)  = 

(P3)  = 

(P4)  $\subseteq$ 

$$(P5) \quad k > 0 \quad \text{---} \boxed{k} \text{---} \boxed{\geq} \text{---} = \text{---} \boxed{\geq} \text{---} \boxed{k} \text{---}$$

$$(P6) \quad k < 0 \quad \text{---} \boxed{k} \text{---} \boxed{\geq} \text{---} = \text{---} \boxed{\leq} \text{---} \boxed{k} \text{---}$$

$$\begin{array}{lcl}
 \text{(antisym)} & \text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\leq} \text{---} \end{array} \bullet \text{---} & \subseteq \text{---} \\
 \text{(spider)} & \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \end{array} \bullet \text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \end{array} \text{---} & = \begin{array}{c} \text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \end{array} \bullet \text{---} \\ \text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \end{array} \bullet \text{---} \end{array} \\
 \text{(direction)} & \text{---} \boxed{\leq} \text{---} \boxed{\geq} \text{---} & = \text{---} \bullet \quad \bullet \text{---}
 \end{array}$$

Definition 3.4 (Graphical Polyhedral Cones Algebra). We call $\mathbf{GPA}_{\vdash}$ the SMT generated by GPA but excluding the affine generator $\vdash$ and its associated equations. This fragment axiomatises polyhedral cones, where no translation is present.

Definition 3.5 (The Free Categories of Polyhedral Diagrams). We call $\mathbf{FreeP}_{\mathbf{GPA}}$ the PROP freely generated by GPA, and $\mathbf{FreeP}_{\mathbf{GPA}_{\vdash}}$ the PROP freely generated by $\mathbf{GPA}_{\vdash}$.

3.3 THE ORDER STRUCTURE OF THE INEQUALITY GENERATOR

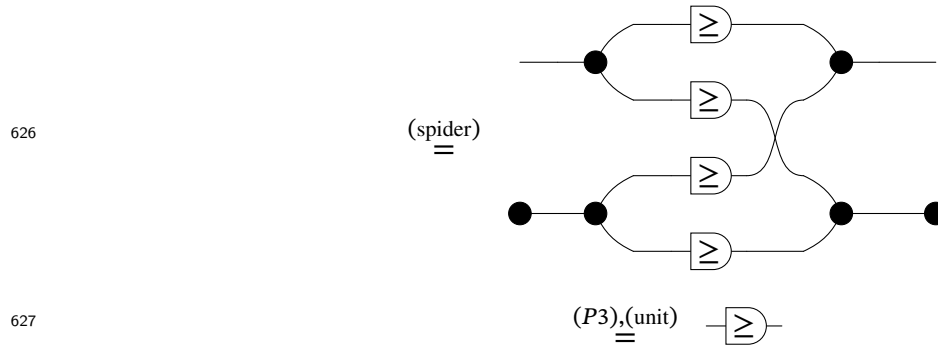
The central computational primitive for reasoning inside this algebra is Fourier–Motzkin Elimination, which projects a polyhedral cone onto a lower-dimensional space by eliminating one variable at a time.

Rescaling and sign are given outright by (P5) and (P6), and the pairwise merge by (spider). What the axioms do not state, and what we need first, is that $\text{---} \boxed{\geq} \text{---}$ deserves its name: reflexivity and transitivity are not among them, and antisymmetry is postulated only as the inclusion $\subseteq$.

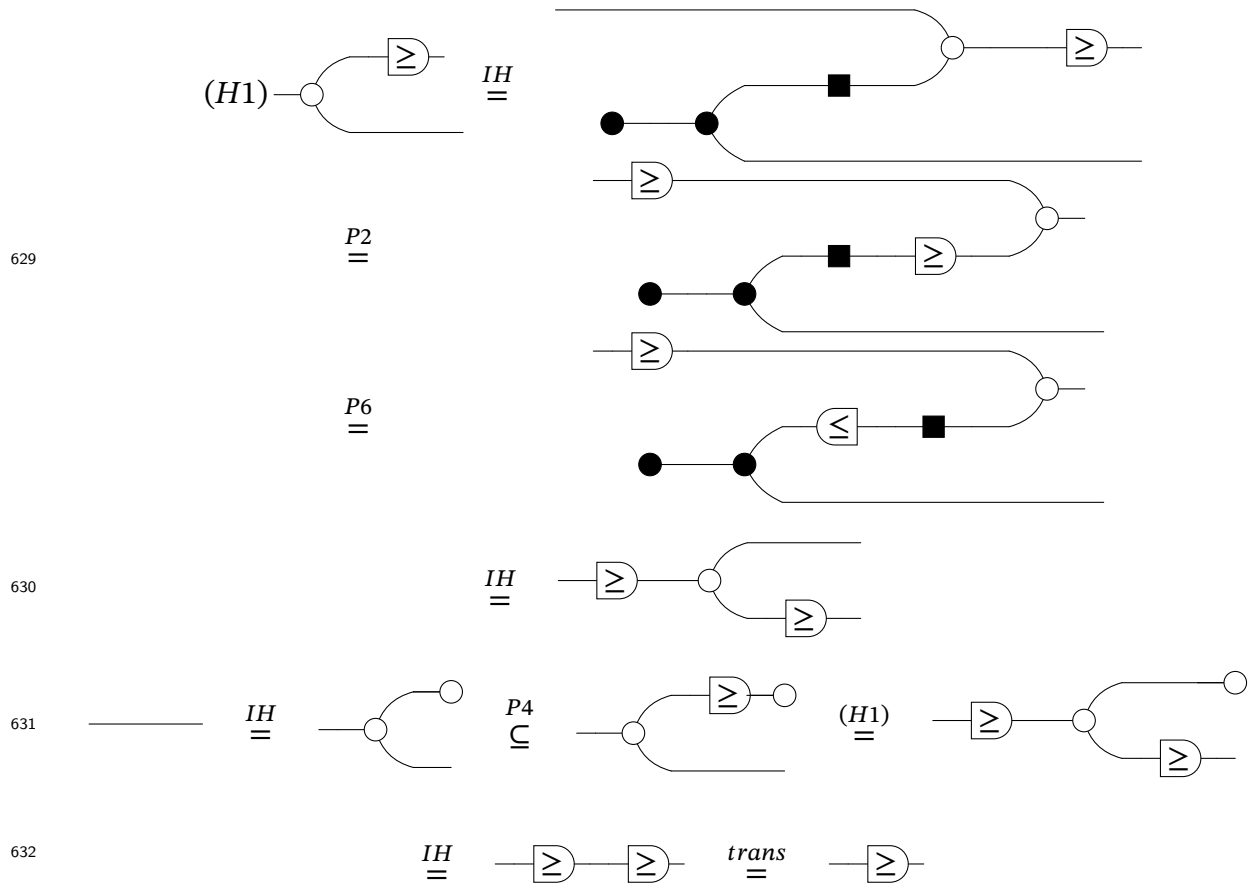
Lemma 3.1 (The generator is an order). In $\mathbf{GPA}_{\vdash}$ the generator $\text{---} \boxed{\geq} \text{---}$ is reflexive, transitive and antisymmetric.

Proof. Transitivity is (spider) with one input fed by the black unit and one output discarded, the plugs being absorbed by (P3) and the black (co)unit laws:

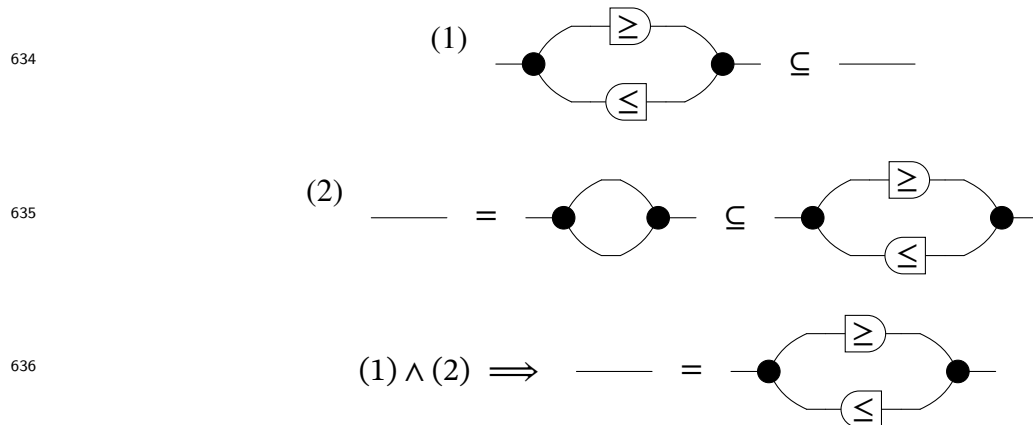
$$\begin{array}{c}
 \text{(unit)} \\
 = \\
 \begin{array}{c} \text{---} \boxed{\geq} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \bullet \text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \end{array} \text{---} \bullet \end{array}
 \end{array}$$



628 For reflexivity we need a previous results that we call $H1$



633 Antisymmetry, the converse of the inclusion (antisym):



638 Since $\boxed{\leq}$ is $\boxed{\geq}$ with an antipode on each side, by (P6) at $k = -1$, every axiom comes
 639 in a mirrored version, and we use both without further comment.

640 The move the axioms leave implicit is the transport of a term across a sum: it is the step that
 641 turns a row of a system into a bound on a single variable, and it is the content of the next lemma.

642 **Lemma 3.2** (Cap form). *In $\text{GPA}_{\perp}$,*

643 (C1)

644 (C2)

645 The first chain reads $x + u \geq v$ iff $v - u \leq x$: an inequality may be applied before or after a
 646 sum indifferently. The second reads $s + u \geq 0$ iff $s \geq -u$, and it is the form we shall use: the
 647 diagram $(\boxed{\geq} \oplus \text{id}); \varepsilon$, whose denotation is $\{(\ell, r) \mid \ell \geq r\}$, we call the *cap form* of $\ell \geq r$, so
 648 that a row $s + u \geq 0$ of a system is the cap form of the bound $s \geq -u$.

649 *Proof.* Bending the second input of the white sum to an output is a bijection on hom-sets, so
 650 the first equation may be compared in bent form; the third follows by commutativity of the sum.
 651 Closing the bent leg with a white cozero then gives the second equation.

652 (C1)

653

654 (C2)

655 □

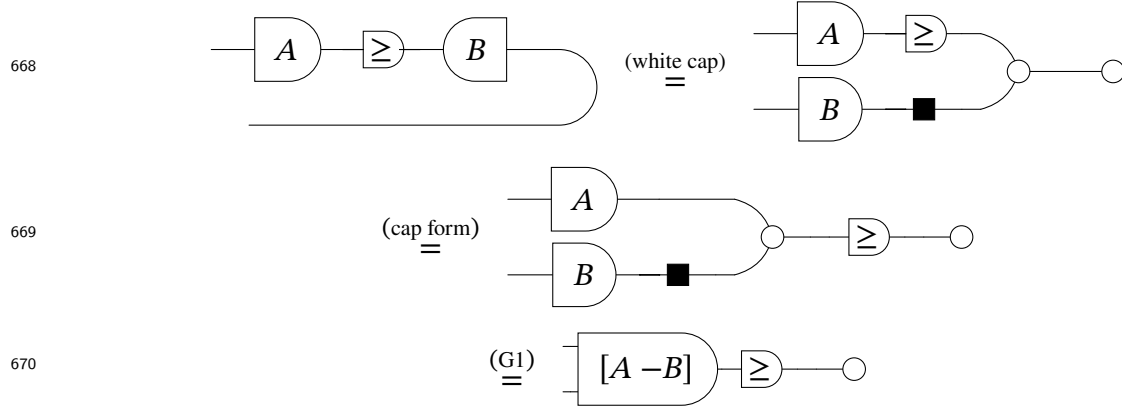
656 The same bending turns a two-sided system $Ax \geq By$ into a homogeneous one. This is
 657 the presentation in which elimination takes place, since it has a single block of rows and no
 658 distinguished side.

659 **Lemma 3.3** (Cone form). *Let $A : n \rightarrow p$ and $B : m \rightarrow p$ be matrices. Bending the m outputs*
 660 *of A ; $\boxed{\geq}$; B^{op} into inputs yields the cone $[A \ -B]; \boxed{\geq}^{\oplus p}; \text{coz}^{\oplus p}$, and unbending recovers*
 661 *the original diagram.*

662

That is, $\exists v. Ax \geq v, v = By$ iff $Ax - By \geq 0$. Rescaling the rows of a cone by positive constants leaves it unchanged, since $Dz \geq 0$ iff $z \geq 0$ for diagonal $D > 0$; this is (P5) again, applied row by row.

Proof. Bending the matrix B exposes a white sum with an antipode on one leg, and the cap form pushes the inequality past it.



□

The third move, the pairwise merge, is (spider). It is the only one of the three that is not local: on its left-hand side a black node forces a single intermediate value below every x_i and above every y_j , and on its right-hand side that value is gone and only the relations $x_i \geq y_j$ survive. The axiom is therefore the elimination of an intermediate variable in the form discussed in the introduction, and the number of rows grows from $a + b$ to ab exactly as it does in the classical algorithm.

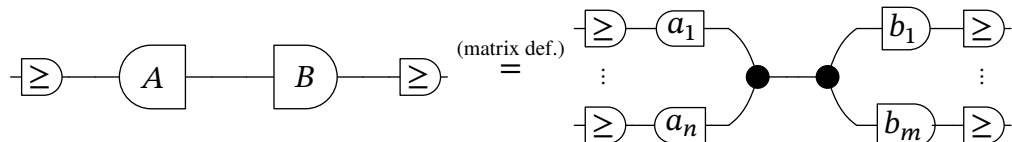
3.4 FOURIER-MOTZKIN ELIMINATION

With the three moves available we can eliminate a single variable. Let $M \in \mathbb{R}^{p \times q}$ have first column a , and split the row indices into $P = \{i \mid a_i > 0\}$, $N = \{j \mid a_j < 0\}$ and $Z = \{i \mid a_i = 0\}$.

Proposition 3.1 (Fourier-Motzkin Elimination). *The following diagrammatic procedure is equivalent to perform the FME algorithm*

$$\begin{array}{c} \geq \\ \hline \end{array} \begin{array}{c} \boxed{A} \\ \hline \end{array} \begin{array}{c} \boxed{B} \\ \hline \end{array} \begin{array}{c} \geq \\ \hline \end{array} = \begin{array}{c} \boxed{A'} \\ \hline \end{array} \begin{array}{c} \geq \\ \hline \end{array} \begin{array}{c} \boxed{B'} \\ \hline \end{array}$$

Proof. First consider one middle wire; ellipses denote parallel repetitions.



For positive coefficients,

$$\begin{aligned}
 \boxed{\geq} - \boxed{d} &\stackrel{\text{inversion lemma}}{=} \boxed{\geq} - \boxed{d^{-1}} \stackrel{(P5)}{=} \boxed{d^{-1}} - \boxed{\geq} \stackrel{\text{inversion lemma}}{=} \boxed{d} - \boxed{\geq} \\
 &\stackrel{(P5)}{=} \boxed{d} - \boxed{\geq} = \boxed{\geq} - \boxed{d}
 \end{aligned}$$

685 and for negative coefficients,

$$\begin{aligned}
 \boxed{\geq} - \boxed{d} &\stackrel{\text{inversion lemma}}{=} \boxed{\geq} - \boxed{d^{-1}} \stackrel{(P6)}{=} \boxed{d^{-1}} - \boxed{\leq} \stackrel{\text{inversion lemma}}{=} \boxed{d} - \boxed{\leq} \\
 &\stackrel{(P6)}{=} \boxed{d} - \boxed{\geq} = \boxed{\leq} - \boxed{d}
 \end{aligned}$$

686 Zero legs detach:

$$\begin{aligned}
 \boxed{\geq} - \boxed{0} &\stackrel{(\text{zero sc.}), (\text{slide})}{=} \boxed{\geq} - \bigcirc - \bullet \\
 \boxed{0} - \boxed{\geq} &\stackrel{(\text{zero sc.})}{=} \bullet - \bigcirc - \boxed{\geq} \\
 \bullet - \bullet &\stackrel{(\text{unit})}{=} \text{---} \quad \bullet - \bullet \stackrel{(\text{unit})}{=} \text{---}
 \end{aligned}$$

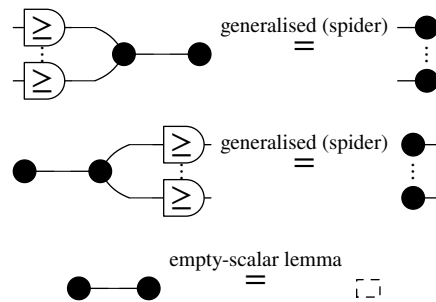
687 Bend the reversed legs:

$$\boxed{\leq} \text{ (bent)} \stackrel{(\text{slide})}{=} \boxed{\geq} \text{ (bent)}$$

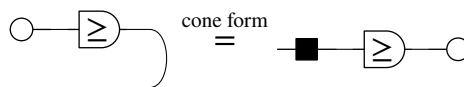
688 The nonzero part now rewrites as follows.

$$\begin{aligned}
 &\begin{array}{c} \boxed{d_1} \\ \vdots \\ \boxed{d_p} \end{array} - \begin{array}{c} \boxed{\geq} \\ \vdots \\ \boxed{\geq} \end{array} \text{---} \bullet \text{---} \bullet \begin{array}{c} \boxed{\geq} \\ \vdots \\ \boxed{\geq} \end{array} \begin{array}{c} \boxed{e_1} \\ \vdots \\ \boxed{e_q} \end{array} \\
 &\stackrel{\text{generalised (spider)}}{=} \begin{array}{c} \boxed{d_1} \\ \vdots \\ \boxed{d_p} \end{array} - \begin{array}{c} \bullet \\ \vdots \\ \bullet \end{array} \begin{array}{c} \boxed{\geq} \\ \vdots \\ \boxed{\geq} \end{array} \begin{array}{c} \bullet \\ \vdots \\ \bullet \end{array} \begin{array}{c} \boxed{e_1} \\ \vdots \\ \boxed{e_q} \end{array} \\
 &\stackrel{\text{inversion lemma}}{=} \begin{array}{c} \boxed{d_1^{-1}} \\ \vdots \\ \boxed{d_p^{-1}} \end{array} - \begin{array}{c} \bullet \\ \vdots \\ \bullet \end{array} \begin{array}{c} \boxed{\geq} \\ \vdots \\ \boxed{\geq} \end{array} \begin{array}{c} \bullet \\ \vdots \\ \bullet \end{array} \begin{array}{c} \boxed{e_1^{-1}} \\ \vdots \\ \boxed{e_q^{-1}} \end{array} \\
 &\stackrel{(\text{matrix def.})}{=} \boxed{H} - \boxed{\geq} - \boxed{K}
 \end{aligned}$$

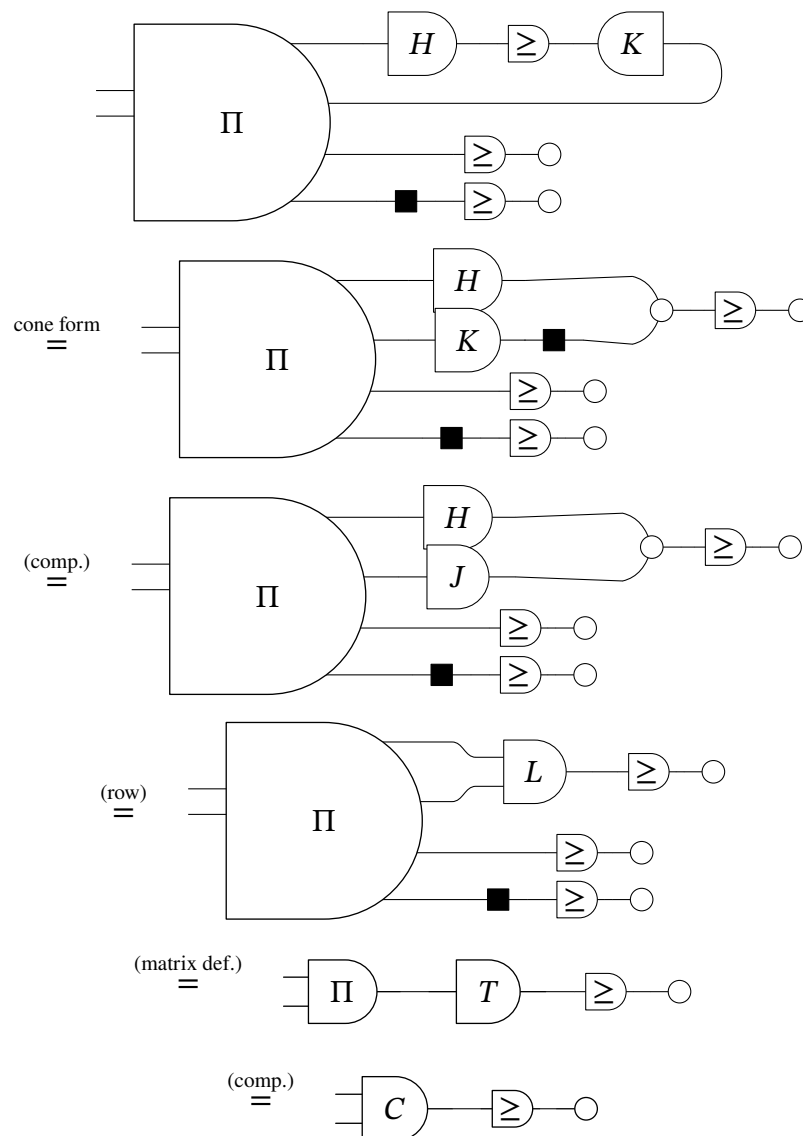
689 For an empty family of bounds,



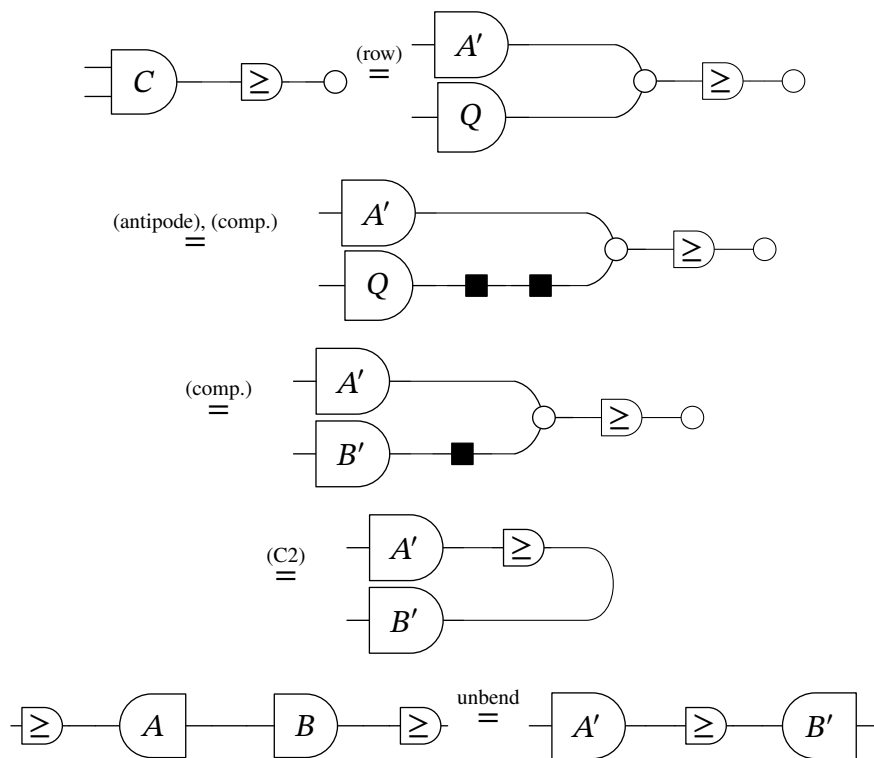
690 Bend the detached output constraints as well:



691 Reattach the zero branches; the permutation restores their boundary positions.

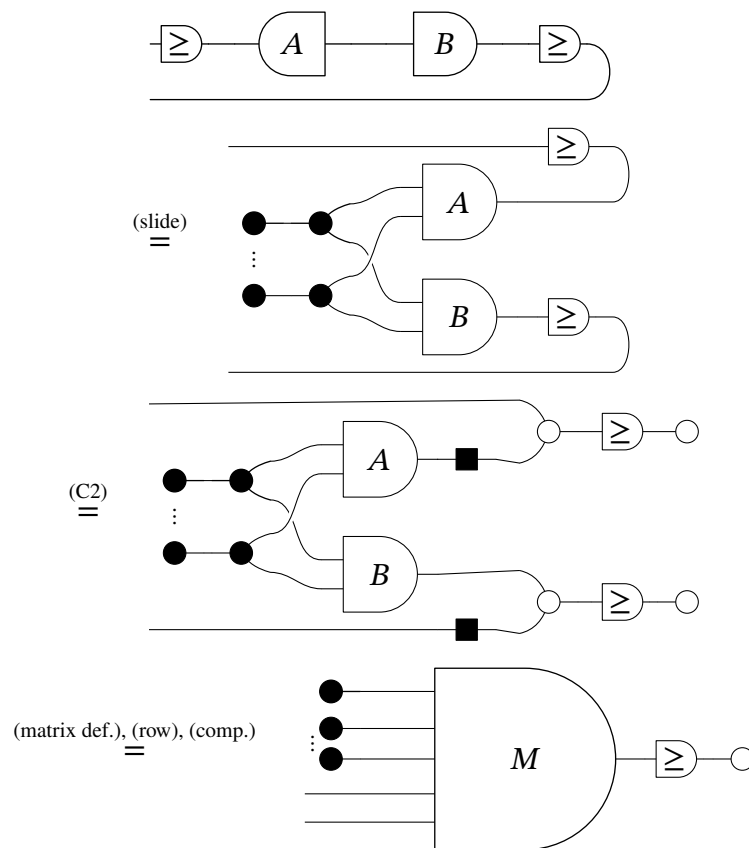


692 Restore the original input–output boundary:



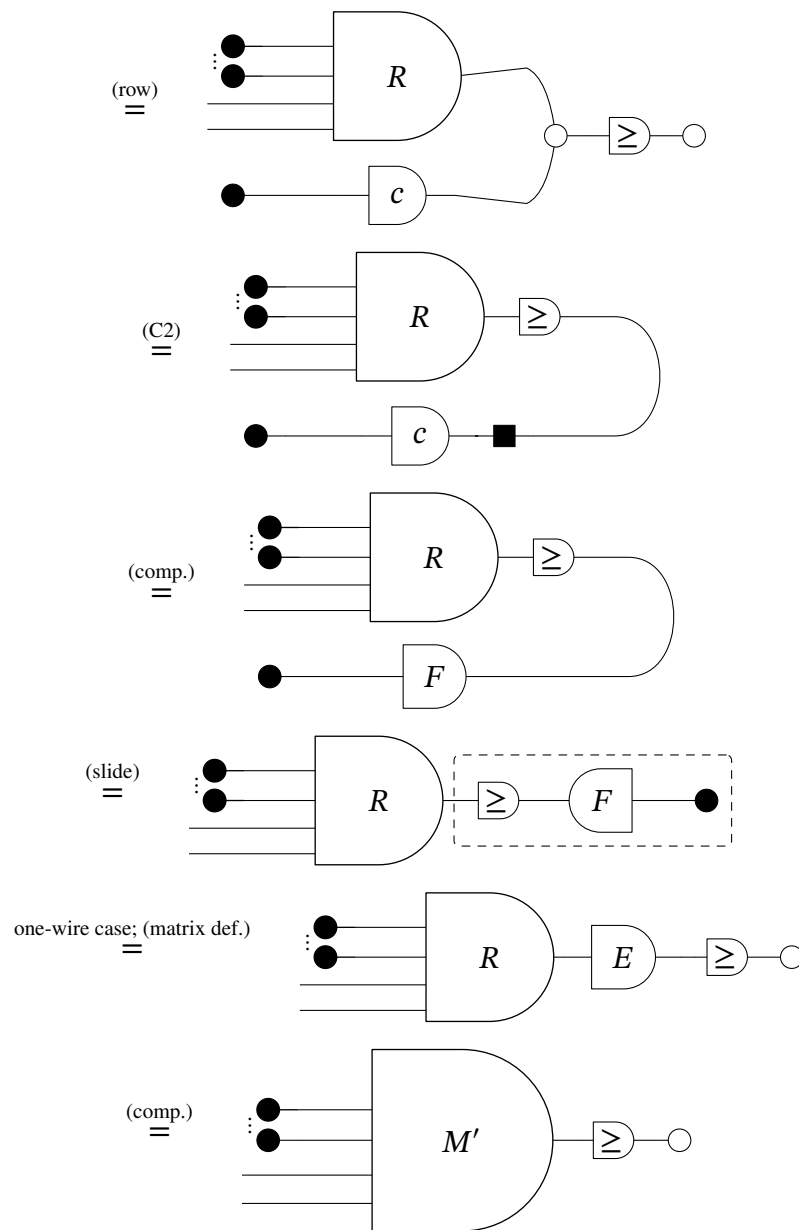
693

Now bend an arbitrary middle bundle and expose its independent sources.



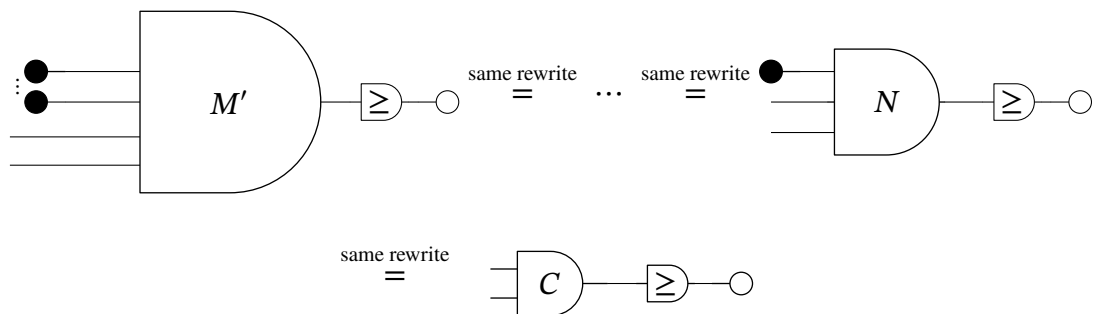
694

Separate one source.



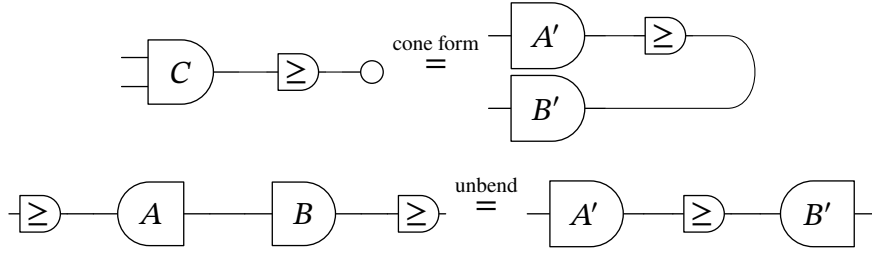
695

Repeat the same local rewrite until no source remains.



696

Apply the boundary-restoration chain above:



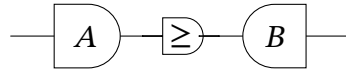
697

□

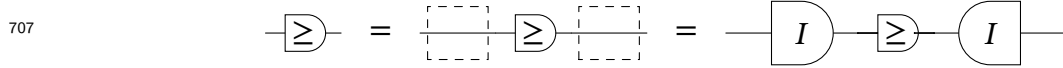
698 A pair (x, y) is related by $\neg \geq$; g ; $\neg \geq$ precisely when there are v, w with $x \geq v$, $(v, w) \in$
 699 $\llbracket g \rrbracket$ and $w \geq y$, so the two bound variables v, w are what the elimination has to remove.

700 With FME available as a diagrammatic operation, we can now prove the normal form theorem
 701 for $\mathbf{GPA}_{\perp}$. We proceed by structural induction, handling generators, composition, and tensor
 702 product in turn, using FME steps to push all inequality generators into a canonical block form.

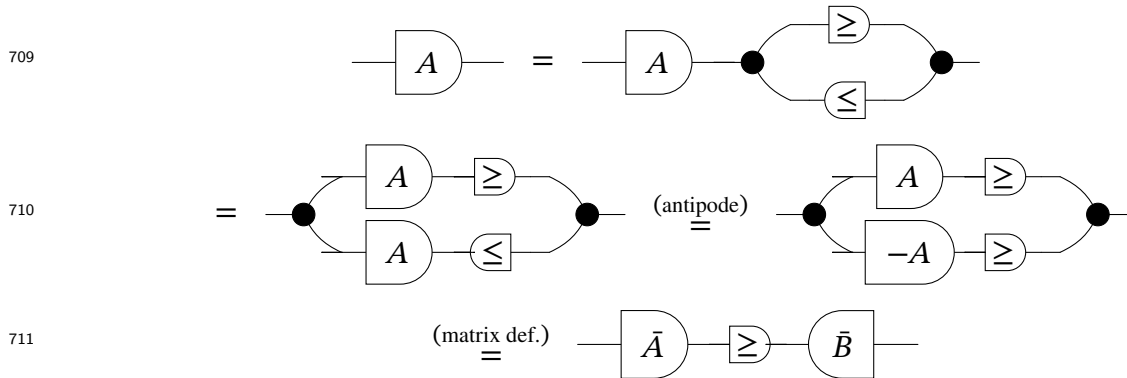
703 **Theorem 3.1** (First Normal Form). *Every diagram in $\mathbf{GPA}_{\perp}$ admits an isomorphic representa-*
 704 *tion, modulo the axioms, of the following form:*



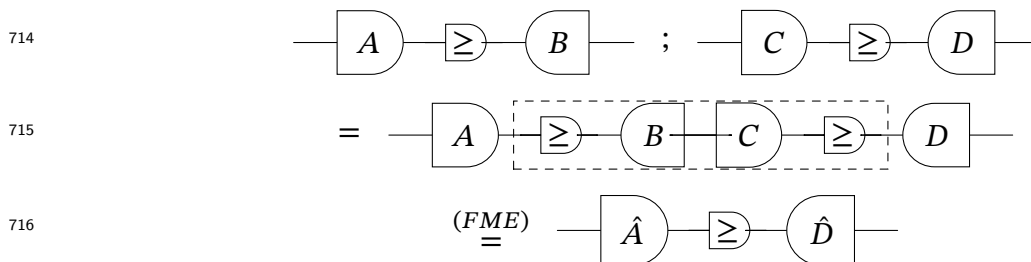
705 *Proof.* We proceed by structural induction. The two base cases are as follows. First, the $\geq$
 706 generator:



708 Second, for any diagram c in $\mathbf{GLA}$:



712 For the inductive step, we consider composition of two diagrams $c; d$ in $\mathbf{GPA}_{\perp}$. By the inductive
 713 hypothesis, both c and d are already in normal form:



716

717 And for the tensor product:

$$\begin{array}{c}
 \text{718} \\
 \text{719}
 \end{array}
 \begin{array}{c}
 \begin{array}{c}
 \text{---} \boxed{A} \text{---} \boxed{\geq} \text{---} \boxed{B} \text{---} \\
 \text{---} \boxed{C} \text{---} \boxed{\geq} \text{---} \boxed{D} \text{---}
 \end{array}
 =
 \begin{array}{c}
 \text{---} \boxed{\bar{A}} \text{---} \boxed{\geq} \text{---} \boxed{\bar{B}} \text{---}
 \end{array}
 \end{array}$$

719 where $\bar{A} = [A \ C]$ and $\bar{B} = [B \ D]$ □

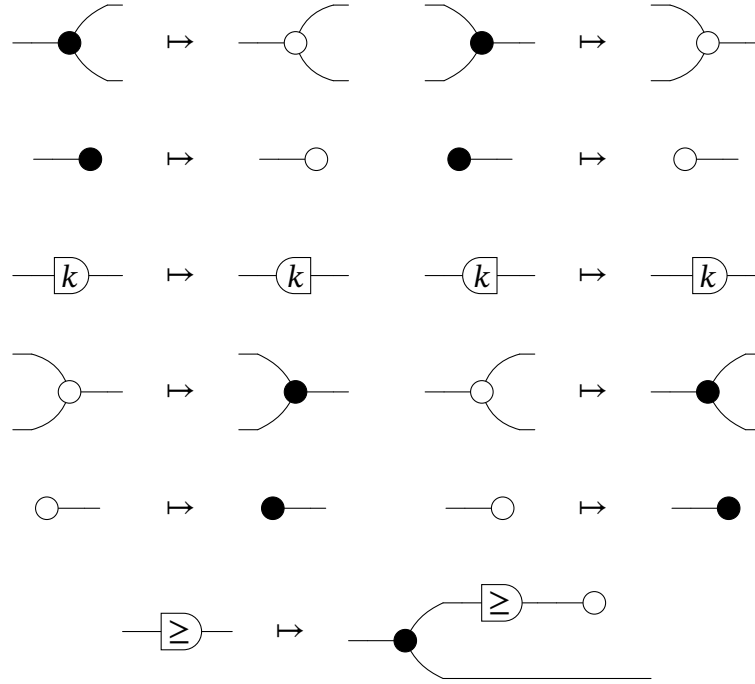
720 Working with a normal form defined up to isomorphism is in principle weaker than requiring
 721 syntactic equality of terms. This choice is motivated by the compact-closed structure, which
 722 provides a canonical bijection $\text{Hom}(A, B) \cong \text{Hom}(\mathbb{1}, A^* \otimes B)$, allowing input wires to be treated
 723 symmetrically with output wires but only up to this isomorphism, not up to equality. As shown
 724 in [BDS21], the same completeness argument can be carried out using strict diagrammatic equal-
 725 ity; however, the derivations become significantly more involved. Since every result established
 726 for the isomorphic normal form carries over to the strict one, the only difference being the need
 727 to bend wires at each step, we adopt the more convenient convention throughout.

728 This idea was already encountered in the treatment of Graphical Quadratic Algebra [Ste+24],
 729 but it is worth pausing to appreciate its significance. The compact-closed structure does not
 730 merely allow a more convenient notation; it enables diagrammatic reasoning to serve as a rigorous
 731 replacement for coordinate-based algebraic manipulation: bending a wire to the left or to the
 732 right does exactly what one would expect. Even more, a statement that would classically require
 733 several lines of matrix algebra, tracking indices, transposing, and substituting, can instead be
 734 verified by a sequence of local wire manipulations, each justified by a single axiom.

735 This first normal form expresses every diagram as a system of inequalities applied to a linear
 736 map. A dual normal form can, in turn, be obtained by applying bijective operator into it and
 737 deriving them, a second normal form. By bijective operator, we mean a fully and faithful
 738 endofunctor in the categories of diagrams

739 This operator is called the polar operator, and plays the role of duality in the polyhedral
 740 setting, exchanging the H -representation (intersection of halfspaces) with the V -representation
 741 (conic hull of generators), in direct analogy with the Farkas lemma (see [BDZ21; BDS21]) for
 742 more details on the operator).

743 **Definition 3.6** (Polar Operator). *The polar operator $-^\circ$ is the involution on generators defined by:*



744 Its action on diagrams is defined functorially by $(a \circ b)^\circ = a^\circ \circ b^\circ$.

745 **Proposition 3.2** (Second Normal Form). *Every diagram $s \in \mathbf{GPA}_\perp$ can be written as:*

$$\left(\text{---} \boxed{A} \text{---} \boxed{\geq} \text{---} \boxed{B} \text{---} \right)^\circ = \text{---} \boxed{A^T} \text{---} \bullet \begin{cases} \boxed{\geq} \text{---} \circ \\ \boxed{B^T} \text{---} \end{cases}$$

746 *Proof.* Apply the polar operator to the first normal form of every diagram $c \in \mathbf{GPA}_\perp$. Since
 747 the polar operator is an involution and acts functorially, the result is again a valid diagram in
 748 $\mathbf{GPA}_\perp$ in the claimed form. $\square$

749 The second normal form expresses every diagram as a conic combination of column vectors,
 750 which is precisely the V -representation of a polyhedral cone. This is the representation that will
 751 drive the completeness proof: two diagrams that are semantically equal must have the same conic
 752 hull, and the axioms of $\mathbf{GPA}_\perp$ are exactly sufficient to witness this equality diagrammatically.

753 3.5 PRESENTATION RESULTS FOR PCREL

754 We now develop the presentation theory for the homogeneous case, proceeding to establish
 755 the three properties that together yield the presentation of PCRel.

756 **Lemma 3.4** (Soundness). *If $s = t$ in $\text{FreeP}_{\mathbf{GPA}_\perp}$, then $[s] = [t]$ in PCRel.*

757 *Proof.* All axioms except those involving the newly introduced $\geq$ generator are already present
 758 in $\mathbf{GAA}$ and therefore verified. It suffices to check that the new axioms preserve the semantics:

759 (P1)

$$\left[\text{---} \boxed{\geq} \text{---} \bullet \right] = \left\{ (x, \binom{y_1}{y_2}) \mid x \geq y_1 \wedge x \geq y_2 \wedge y_1 = y_2 \right\}$$

760

$$\subseteq \left\{ (x, \binom{y_1}{y_2}) \mid x \geq y_1 \wedge x \geq y_2 \right\} = \left[\text{---} \bullet \begin{array}{l} \boxed{\geq} \\ \boxed{\geq} \end{array} \right]$$

761 (P2)

$$\left[\begin{array}{l} \text{---} \end{array} \bigcirc \text{---} \boxed{\geq} \text{---} \right] = \left\{ \left(\binom{x_1}{x_2}, y \right) \mid x_1 + x_2 \geq y \right\} = \left\{ \left(\binom{x_1}{x_2}, y \right) \mid \exists c, d. x_1 + x_2 \geq c + d \wedge c + d = y \right\}$$

762

$$= \left\{ \left(\binom{x_1}{x_2}, y \right) \mid \exists \hat{c}, \hat{d}. x_1 \geq \hat{c} \wedge x_2 \geq \hat{d} \wedge \hat{c} + \hat{d} = y \right\} = \left[\begin{array}{l} \boxed{\geq} \\ \boxed{\geq} \end{array} \bigcirc \text{---} \right]$$

763 (P3)

$$\left[\text{---} \boxed{\geq} \text{---} \bullet \right] = \{ (x, \bullet) \mid \exists c. x \geq c \} \underset{\text{tautology}}{=} \{ (x, \bullet) \mid x \in \mathbb{R} \} = \left[\text{---} \bullet \right]$$

764 (P4)

$$\left[\bigcirc \text{---} \right] = \{ (x, \bullet) \mid x = 0 \} \subseteq \{ (x, \bullet) \mid x \geq 0 \} = \left[\bigcirc \text{---} \boxed{\geq} \text{---} \right]$$

765 (P5), (P6)

766 For $k > 0$:

$$\left[\text{---} \boxed{k} \text{---} \boxed{\geq} \text{---} \right] = \{ (x, y) \mid kx \geq y \wedge k > 0 \} = \left\{ (x, y) \mid x \geq \frac{y}{k} \wedge k > 0 \right\}$$

767

$$= \{ (x, y) \mid \exists c. x \geq c \wedge kc = y \wedge k > 0 \} = \left[\text{---} \boxed{\geq} \text{---} \boxed{k} \text{---} \right]$$

768

For $k < 0$: the same computation, dividing by the negative k flips the inequality, so

$$\left[\text{---} \boxed{k} \text{---} \boxed{\geq} \text{---} \right] = \left[\text{---} \boxed{\leq} \text{---} \boxed{k} \text{---} \right]$$

769 (ANTISYM)

$$\begin{aligned}
 & \left[\text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\leq} \text{---} \end{array} \bullet \text{---} \right] = \{(x, y) \mid x \geq y \wedge x \leq y\} \\
 & = \{(x, y) \mid x = y, x, y \in \mathbb{R}\} \subseteq \{(x, y) \mid x = y, x, y \in \mathbb{R}\} = \left[\text{---} \right]
 \end{aligned}$$

771 (SPIDER)

$$\begin{aligned}
 & \left[\begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \end{array} \bullet \text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \end{array} \right] = \left\{ \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right) \mid \exists z. x_1 \geq z \wedge x_2 \geq z \wedge z \geq y_1 \wedge z \geq y_2 \right\} \\
 & = \left\{ \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right) \mid x_1 \geq y_1 \wedge x_1 \geq y_2 \wedge x_2 \geq y_1 \wedge x_2 \geq y_2 \right\} = \left[\begin{array}{c} \text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \end{array} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \end{array} \bullet \text{---} \\ \text{---} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \end{array} \bullet \begin{array}{c} \text{---} \boxed{\geq} \text{---} \\ \text{---} \boxed{\geq} \text{---} \end{array} \bullet \text{---} \end{array} \right]
 \end{aligned}$$

773 (DIRECTION)

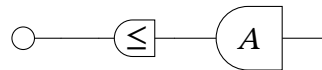
$$\left[\text{---} \boxed{\leq} \text{---} \boxed{\geq} \text{---} \right] = \{(x, y) \mid \exists z. z \geq y \wedge z \geq x\} = \{(x, y) \mid x, y \in \mathbb{R}\} = \left[\text{---} \bullet \quad \bullet \text{---} \right]$$

774

□

775 **Lemma 3.5** (Expressivity). *For every $f : m \rightarrow n$ in $\mathbf{PCRel}$ there exists a Σ -term $s : m \rightarrow n$*
 776 *in $\mathbf{Hom}_{\mathbf{FreeP}_{\mathbf{GPA}} \vdash} (m, n)$ such that $[s] = f$.*

777 *Proof.* By Definition 3.1, every polyhedral cone has the form $\{x \mid Ax \geq 0, x \in \mathbb{R}^n\}$. This is
 778 precisely the interpretation of the diagram:

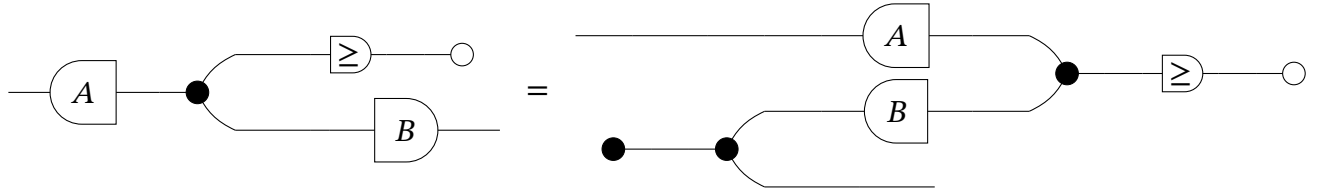


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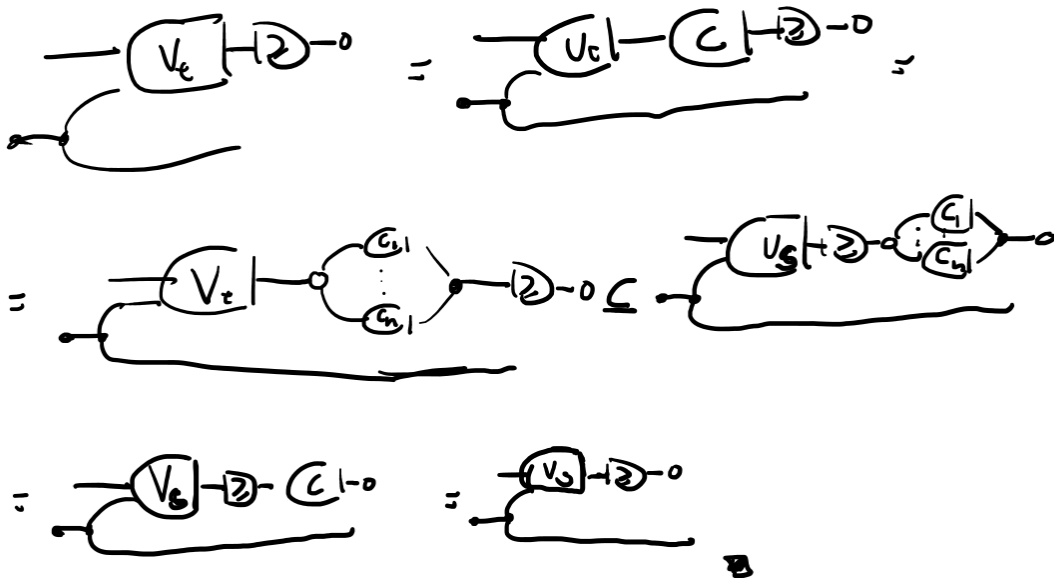
□

780 **Lemma 3.6** (Completeness). *If $[s] = [t]$ in $\mathbf{PCRel}$, then $s = t$ in $\mathbf{FreeP}_{\mathbf{GPA}} \vdash$.*

781 *Proof.* By the second normal form, we write s and t in V -representation form:



782 The semantics of such a diagram is the conic hull of the column vectors of V . Since $[s] = [t]$
 783 implies in particular $[s] \subseteq [t]$, the conic combinations of V_t are a subset of those of V_s . Hence
 784 there exists a non-negative matrix C such that $CV_t = V_s$. The diagrammatic witness is:



785 The inclusion $[s] \subseteq [t]$ follows by the same argument, yielding equality. □

786 **Theorem 3.2.** PCRel is presented by $\text{FreeP}_{\text{GPA}_{\perp}}$.

787 *Proof.* Soundness, expressivity, and completeness were established by the preceding three lem-
 788 mas. Together they imply that the canonical PROP morphism is an isomorphism. □

789 This settles the homogeneous case: every polyhedral cone is diagrammatically expressible,
 790 and two diagrams denote the same cone exactly when the axioms can prove them equal. Cones
 791 are, however, a narrow class, they always contain the origin, so they cannot represent an arbitrary
 792 feasible region or, later, an arbitrary linear program. The next step lifts this result from cones to
 793 general polyhedra, replacing the homogeneous inequalities $Ax \geq 0$ with the affine $Ax \geq b$, and
 794 the machinery developed above will carry over almost unchanged.

795 3.6 SEMANTICS: THE PROP POLYREL OF POLYHEDRA

796 Now we start the presentation of the broader class of polyhedral relations by recalling the
797 definition of a polyhedron.

798 **Definition 3.7.** *A set $P \subseteq \mathbb{R}^n$ is called a polyhedron if there exist $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ such*
799 *that*

$$P = \{x \in \mathbb{R}^n \mid Ax \geq b\}.$$

800 Note that a polyhedral cone is a special case of a polyhedron with $b = 0$. The distinction
801 is algebraically significant: cones are closed under non-negative scaling and admit a purely
802 homogeneous representation, while general polyhedra may be translated. This distinction is
803 reflected in the two graphical algebras we now define, where $\mathbf{GPA}_{\vdash}$ handles the homogeneous
804 case and $\mathbf{GPA}$ extends it with an affine generator.

805 In that light, one can define the category of polyhedral relations as:

806 **Definition 3.8** (The prop $\mathbf{PolyRel}$ of polyhedra). *Let k be an ordered field. The prop $\mathbf{PolyRel}_k$*
807 *is defined as follows:*

- 808 • *Objects are natural numbers $n \in \mathbb{N}$, interpreted as the space k^n .*
- 809 • *A morphism $n \rightarrow m$ is a polyhedron $R \subseteq k^n \times k^m$ for which there exist $p \in \mathbb{N}$, $A \in$*
810 *$k^{p \times (n+m)}$, and $b \in k^p$ such that*

$$R = \left\{ (x, y) \in k^n \times k^m \mid A \begin{pmatrix} x \\ y \end{pmatrix} + b \geq 0 \right\}.$$

- 811 • *Composition is relational composition.*
- 812 • *The monoidal product on objects is addition $n \otimes m := n + m$, and on morphisms is the*
813 *Cartesian product.*

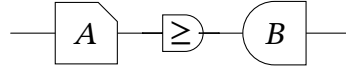
814 Notice how, the same way that $\mathbf{PCRel}$ offered us a more relational way to think about
815 Polyhedral Cones, so does $\mathbf{PolyRel}$. Indeed, one may think of polyhedra as closed spaces in
816 some geometry that respects some linear constraints. We argue, however, that this definition is
817 overly convoluted, and a more natural way to denote such objects is to think in terms of relations
818 R , such that $xRy \iff Ax + b \geq By + c$ for some matrices A, B and vectors b, c . Once again,
819 we move from a set-theoretic heavy description to a more logical one.

820 3.7 SYNTAX: THE ALGEBRA GPA

821 We now extend the theory from polyhedral cones to general polyhedra by reintroducing the
822 affine generator $\vdash$. Geometrically, this corresponds to allowing translations: a polyhedron is

a translated polyhedral cone, and the generator $\vdash$ provides the syntactic means to encode the right-hand side vector b in $Ax \geq b$. The normal form, soundness, and expressivity arguments all extend smoothly from the homogeneous case.

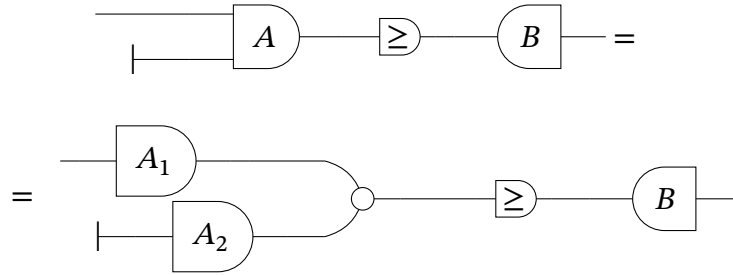
Theorem 3.3 (Normal Form). *For every diagram c in $\mathbf{GPA}$, there exists a diagram d such that $c = d$ modulo the axioms and d is of the form:*



Proof. The derivation proceeds directly. Let c be a diagram of $\mathbf{GPA}$. Then we can collect all $\vdash$ on the left such that c' is a diagram of $\mathbf{GPA}_{\vdash}$



From Theorem 3.1 we know that c' admits a normal form in terms of matrices A and B . Breaking down the matrix A in columns we get:



thus the thesis. □

3.8 PRESENTATION RESULTS FOR POLYREL

Lemma 3.7 (Soundness). *If $s = t$ in $\text{FreeP}_{\mathbf{GPA}}$, then $[s] = [t]$ in PolyRel .*

Proof. This follows immediately from the soundness of $\mathbf{PCRel}$. Since the generator $\vdash$ introduces no new axioms beyond those already verified for $\mathbf{PCRel}$ and $\mathbf{GAA}$, and both of those theories are sound in PolyRel , soundness carries over directly. □

Lemma 3.8 (Expressivity). *For every $f : m \rightarrow n$ in PolyRel there exists a Σ -term $s : m \rightarrow n$ in $\text{Hom}_{\text{FreeP}_{\mathbf{GPA}}}(m, n)$ such that $[s] = f$.*

Proof. By Definition 3.7, every polyhedron has the form $\{x \mid Ax \geq b\}$. Appealing to this representation directly, the polyhedron is expressed as the interpretation of the following diagram, where the generator $\vdash$ encodes the right-hand side vector b :

$$\left[\vdash \boxed{b} \multimap \boxed{A} \vdash \right] = \{ (b, x) \mid Ax \geq b, x \in \mathbb{R}^n \}$$

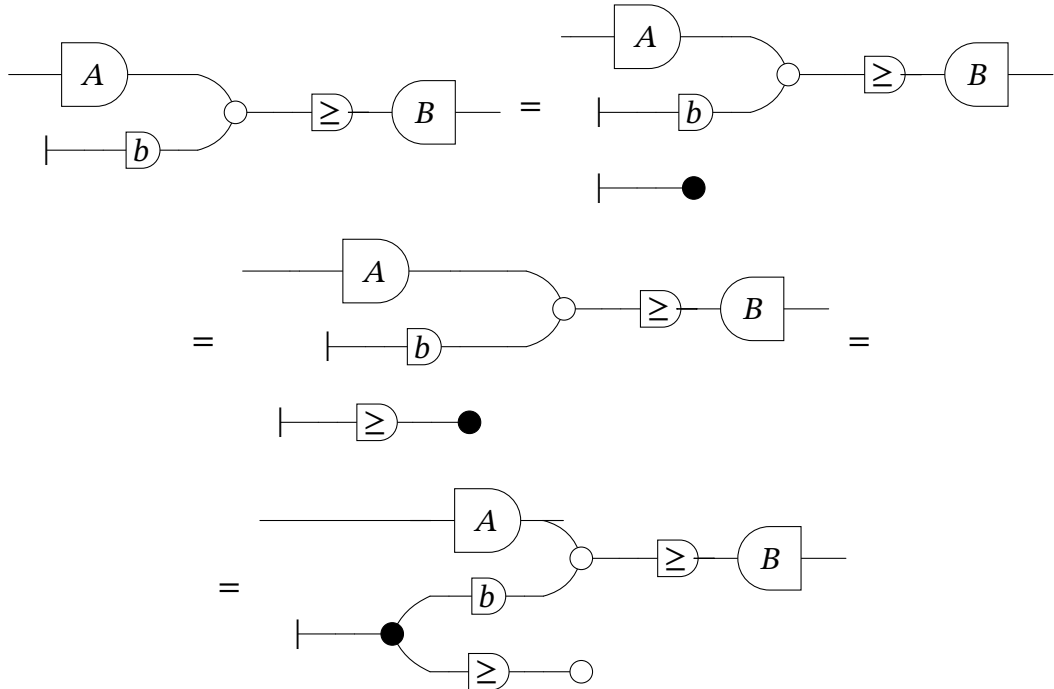
843

□

844 **Lemma 3.9** (Completeness). *If $[s] = [t]$ in **PolyRel**, then $s = t$ in $\text{FreeP}_{\text{GPA}}$.*

845 *Proof.* Every polyhedron P can be homogenised by introducing a slack variable, reducing the
 846 affine case to the conic one $P^H = \{(x, y) \in k^{n+1} \mid Ax + by \geq 0, y \geq 0\}$ which admits a normal form
 847 (Theorem 3.1). Under this homogenisation, for each two diagrams $s, t \in \text{GPA}$ with $[s] = [t]$
 848 in **PolyRel**, there are s', t' in $\text{GPA}_{\perp}$ such that $[s'] = [s]^H, [t'] = [t]^H$. Since two polyhedrals
 849 P, Q are equal iff $P^H = Q^H$, then s and t have equal semantics in **PCRel**. Completeness of
 850 $\text{GPA}_{\perp}$ then gives equality of the homogenised diagrams, and de-homogenising recovers equality
 851 of s and t in $\text{FreeP}_{\text{GPA}}$.

852 Diagrammatically this goes as follows:



853 Whereas the interpretation of the last diagram is $\{(x, y, z) \in k^m \mid Ax + by \geq Bz, y \geq 0\}$,
 854 which is the same as P^H . Therefore, the semantics of s' is that of the homogenisation of $[s]$ and
 855 follows the above argument. □

856 **Theorem 3.4.** ***PolyRel** is presented by $\text{FreeP}_{\text{GPA}}$.*

857 *Proof.* By the preceding three lemmas. □

858 This is the result the rest of the thesis leans on: every polyhedral relation, not just the conic
859 ones from earlier, has a diagrammatic representative, and semantic equality between polyhedra
860 can always be certified by a finite rewriting sequence in **GPA**. It is this presentation, together
861 with its normal form, that Chapter 4 builds on when it adds a single new generator for the objective
862 function.

863 4 THE LINEAR PROGRAMMING CATEGORY

864 The theory developed so far has extended the categorical framework by adding a new class of
 865 morphisms. We now introduce the final and most expressive theory of the thesis: a graphical
 866 algebra for parametric linear programs. Linear programs combine both the inequality structure
 867 of polyhedra and the notion of an objective function, and their value functions are precisely the
 868 polyhedral convex functions. The central result of this section is the presentation of a complete
 869 monoidal theory for right-hand side parametric linear programming.

870 4.1 SEMANTICS: THE PROP LPREL

871 We begin with several classical results that establish the tight relationship between piecewise
 872 affine convex functions, polyhedral functions, and parametric linear programs. These will serve
 873 as the semantic foundation for the graphical algebra introduced in the following section, and they
 874 justify describing the morphisms of this chapter's semantic category, **LPRel**, interchangeably
 875 as parametric linear programs or as polyhedral functions.

876 **Definition 4.1** (Piecewise Affine Convex Function). *A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is called piecewise*
 877 *affine convex if there exist finitely many affine functions $a_i^\top x + b_i$, $i = 1, \dots, k$, such that*

$$f(x) = \max_{i=1, \dots, k} \{a_i^\top x + b_i\}.$$

878 **Definition 4.2** (Polyhedral Function). *A function $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is polyhedral if its*
 879 *epigraph*

$$\text{epi}(f) := \{(x, t) \mid t \geq f(x)\}$$

880 *is a polyhedron.*

881 **Definition 4.3** (RHS Parametric Linear Program). *A right-hand side parametric linear program*
 882 *has value function*

$$v(b) := \inf_x \{c^\top x \mid Ax \geq b\}.$$

883 The following two propositions establish that all three notions above are equivalent descrip-
 884 tions of the same class of functions. This equivalence is the semantic cornerstone of the section:
 885 it is what allows us to treat polyhedral functions and parametric linear programs as morphisms in
 886 the same category.

887 **Proposition 4.1.** *A function is piecewise affine convex if and only if it is a polyhedral function.*

888 *Proof.* ($\Rightarrow$) Let $f(x) = \max_{i=1, \dots, k} (a_i^\top x + b_i)$. Then

$$\text{epi}(f) = \bigcap_{i=1}^k \{(x, t) \mid t \geq a_i^\top x + b_i\},$$

889 which is an intersection of finitely many halfspaces, hence a polyhedron.

890 ($\Leftarrow$) If f is polyhedral, its epigraph is a polyhedron, so there exist finitely many affine
891 inequalities $t \geq a_i^\top x + b_i$ such that $f(x) = \max_{i=1,\dots,k}(a_i^\top x + b_i)$. Thus f is piecewise affine
892 and convex. $\square$

893 **Proposition 4.2.** *A function $v : \mathbb{R}^m \rightarrow \mathbb{R} \cup \{+\infty\}$ is piecewise affine convex if and only if it is*
894 *the value function of a right-hand side parametric linear program.*

895 *Proof.* ($\Rightarrow$) Let $v(b) = \max_{i=1,\dots,k}(y^i)^\top b$ and define $Y := \text{conv}\{y^1, \dots, y^k\}$. Then $v(b) =$
896 $\max_{y \in Y} b^\top y$. Since Y is a polyhedron admitting the representation $Y = \{y \mid A^\top y = c, y \geq 0\}$,
897 linear programming duality gives $v(b) = \inf\{c^\top x \mid Ax \geq b\}$.

898 ($\Leftarrow$) Let $v(b) = \inf\{c^\top x \mid Ax \geq b\}$. Dualising, $v(b) = \sup\{b^\top y \mid A^\top y = c, y \geq 0\}$. Since
899 the feasible set $Y = \{y \mid A^\top y = c, y \geq 0\}$ is a polyhedron, the supremum is attained at its extreme
900 points $y^1, \dots, y^k$, yielding $v(b) = \max_{i=1,\dots,k}(y^i)^\top b$. Thus v is piecewise affine convex. $\square$

901 The following proposition is the key technical lemma for the completeness proof of GLP.
902 It states that two parametric linear programs with the same value function must share the same
903 epigraph polyhedron.

904 **Proposition 4.3.** *Let*

$$v_i(y) := \inf\{1^\top x \mid A_i x \geq B_i y\}, \quad i = 1, 2,$$

905 *where $A_i \in \mathbb{R}^{m_i \times n_i}$ and $B_i \in \mathbb{R}^{m_i \times k}$. If $v_1(y) = v_2(y)$ for all $y \in \mathbb{R}^k$, then both programs admit*
906 *a common representation through the same epigraph polyhedron:*

$$v_i(y) = \inf\{t \mid (y, t) \in E\}, \quad E := \text{epi}(v_1) = \text{epi}(v_2).$$

907 *Proof.* For each $i = 1, 2$, the epigraph of v_i satisfies

$$(y, t) \in \text{epi}(v_i) \iff \exists x : A_i x \geq B_i y, 1^\top x \leq t,$$

908 so $\text{epi}(v_i)$ is the projection onto (y, t) of the polyhedron $P_i = \{(x, y, t) \mid A_i x \geq B_i y, 1^\top x \leq t\}$.
909 Projections of polyhedra are polyhedra, so each $\text{epi}(v_i)$ is a polyhedron. Since $v_1 = v_2$ pointwise,
910 we have $\text{epi}(v_1) = \text{epi}(v_2) =: E$. Setting $E = \{(y, t) \mid Cy + dt \geq 0\}$, we obtain $v_i(y) = \inf\{t \mid$
911 $Cy + dt \geq 0\}$ for both $i = 1, 2$. $\square$

912 With the equivalences above in place, we now define the semantic category of this chapter.
913 **LPRel** describes morphisms as parametric linear programs; by the propositions above, these
914 coincide with polyhedral functions, and we will use both descriptions interchangeably according
915 to which is more convenient.

916 **Definition 4.4** (The prop LPRel of Linear Programs). *The prop LPRel is defined as follows:*

- Objects are natural numbers $n \in \mathbb{N}$, interpreted as the space $\mathbb{R}^n$.
- A morphism $n \rightarrow m$ is a right-hand side parametric linear program with value function

$$(x, y) \mapsto \inf_z \{c^\top z \mid Az \geq B[x \ y]^\top + b\},$$

where $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$.

- Composition is given by minimisation over the shared variable.
- The monoidal product on objects is addition $n \otimes m := n + m$, and on morphisms is direct sum.

4.2 SYNTAX: THE ALGEBRA GLP

With the semantic equivalences established, we now introduce the final graphical algebra of the thesis. The algebra **GLP** extends **GPA** with a single new generator whose semantics encodes the linear objective function $c^\top x$. This new generator lifts **GPA** from polyhedral constraints to full linear programs by attaching a linear cost to the feasible region.

Definition 4.5 (GLP). We call **GLP** the symmetric monoidal theory obtained by extending **GPA** with the following generator:

$$\llbracket \square - \rrbracket = (\bullet, x) \mapsto x$$

together with the following axioms governing its interaction with the existing generators:

$$\begin{aligned}
 \text{(linearity)} \quad & \begin{array}{c} \square - \\ \square - \end{array} = \begin{array}{c} \square - \end{array} \circ \begin{array}{c} \text{ } \end{array} \\
 \text{(zeroeq)} \quad & \begin{array}{c} \square - \end{array} \circ \begin{array}{c} \text{ } \end{array} = \begin{array}{c} \square - \end{array} \\
 \text{(inf)} \quad & \begin{array}{c} \square - \end{array} \circ \begin{array}{c} \square - \end{array} = \begin{array}{c} \square - \end{array}
 \end{aligned}$$

The axioms represent expected properties of the linear cost function.

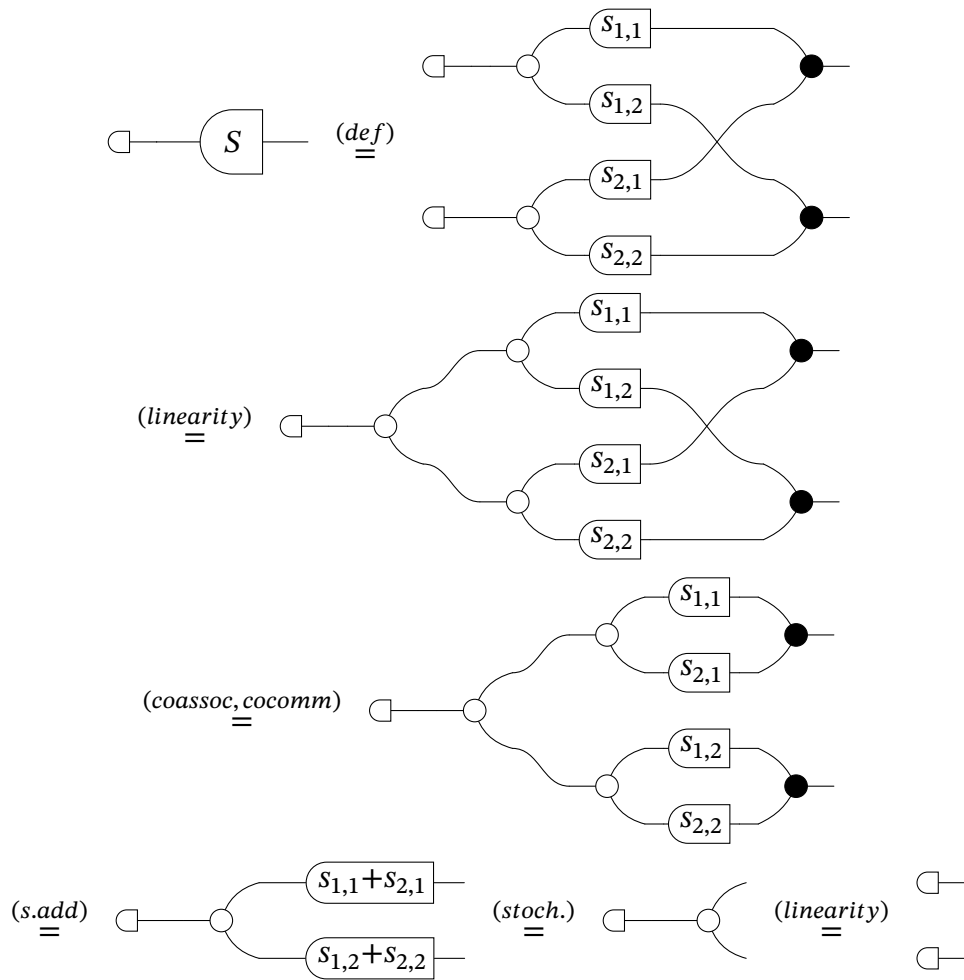
- (linearity) express the linear property $f(x) + f(y) = f(x + y)$. Observe that f is the interpretation of the $\square -$ and the sum denotes the vertical composition; composing two $\square -$ vertically (adding) is the same as adding the wires in a single $\square -$.
- (zeroeq) is the tautological fact that $\inf 0 = 0$
- (inf) is the epigraph property, this is, the fact that the infimum of the epigraph of a function, is the function itself $g(y) = \inf\{x : (y, x) \in \text{epi}(f)\} = \inf_{x \geq f(y)} x = f(y)$.

938 From this simple axioms, we get immediatly another core property of linear programs, namely,
 939 that they are invariant to stochastic matrices.

940 **Remark 4.1** (Invariance under column-stochastic matrices). *The linearity axiom encodes the*
 941 *fundamental invariance of linear programs under column-stochastic reweightings of the objective:*
 942 *for any $n \times m$ matrix S whose columns each sum to one,*

$$\text{---} \boxed{S} \text{---} = \text{---}$$

943 *The proof is diagrammatic, we display the proof for a 2×2 matrix S with $s_{1,1} + s_{2,1} = s_{1,2} + s_{2,2} = 1$;*
 944 *every step is an instance of an axiom scheme that applies wire-by-wire, so the same chain of*
 945 *rewrites goes through verbatim for arbitrary $n \times m$.*



946 This is a pattern present in every algebra discussed so far, where a particular class of matrices
 947 is invariant to one of its core generators.¹ Here, the new class of matrices that interact with

¹In the digrammatic calculi of Linear subspaces (GLA) one can show that surjective matrices are invariant to $\bullet$ — which is a very important feature in the completeness of the language itself. The same occurs in GPA, where non-negative matrices slide through the $\text{---} \boxed{\geq} \text{---}$. Other works have spotted the same pattern, such as [Ste+24] which

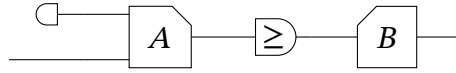
the new generator $\square$ — is that of stochastic matrices, agreeing with what one would expect when discussing linear problems.

With the SMT above we now define the complete category.

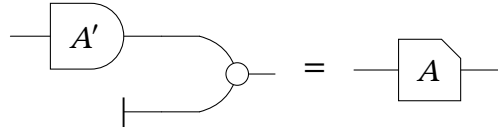
Definition 4.6. We call $\text{FreeP}_{\text{GLP}}$ the PROP freely generated by GLP.

The normal form theorem for GLP extends the one for GPA: every diagram can be reduced to a GPA block encoding the constraints, composed with the new generator encoding the objective. The proof proceeds by structural induction, using the existing normal form for GPA as the inductive base.

Theorem 4.1 (Normal Form). For every diagram $c \in \text{GLP}$, there exists a diagram c' in normal form such that $c = c'$:

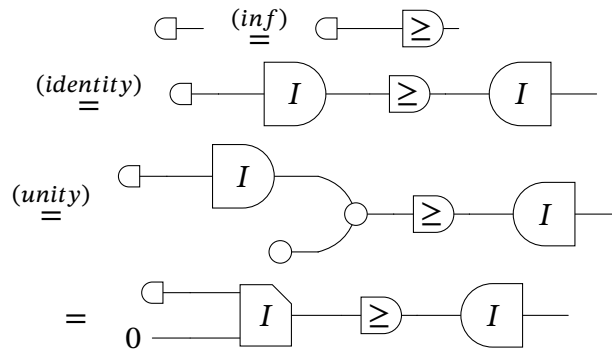


With A and b being an affine transformation. Namely:



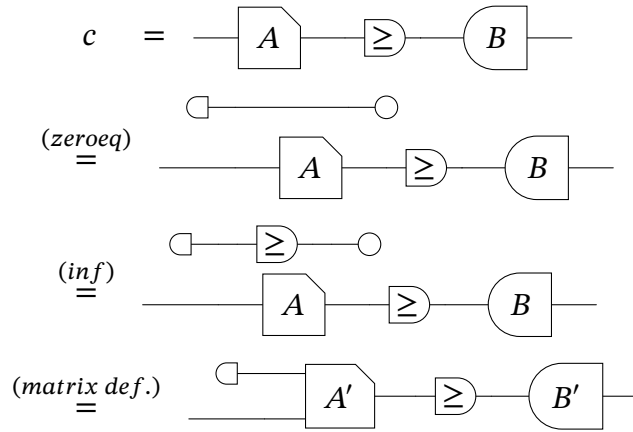
Proof. The proof is by structural induction. First we show the two base cases:

• The generator $\square$ —

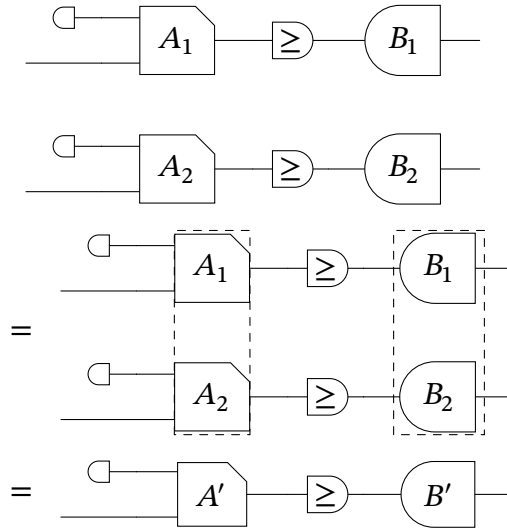


• A diagram $c \in \text{GPA}$

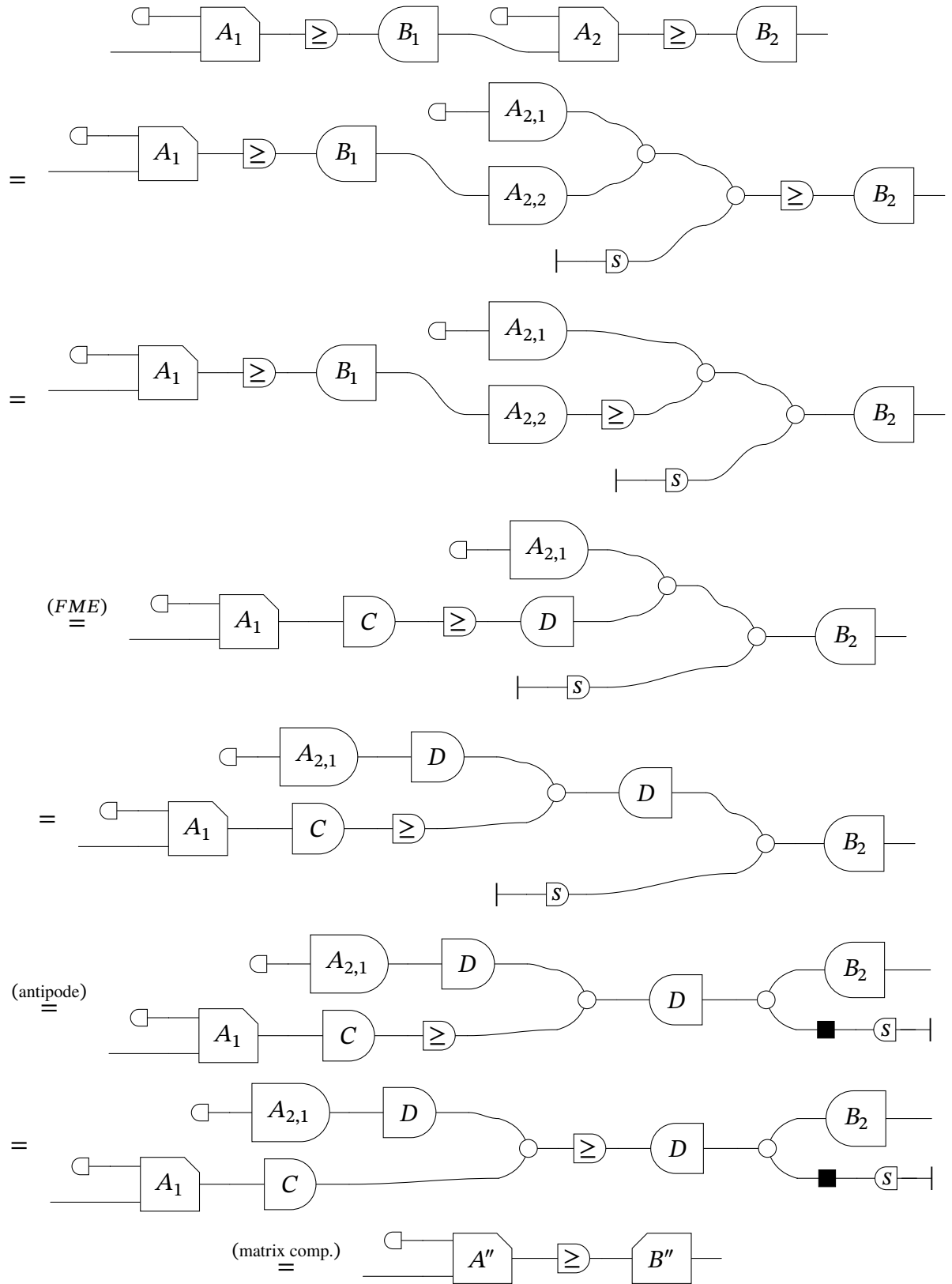
shows that a generator representing the l_2 norm is invariant to orthogonal matrices (also are normal distributed random variables).



962 For the inductive step, we verify that for any two diagrams $c, d \in \mathbf{GLP}$ satisfying the inductive
 963 hypothesis, both horizontal and vertical composition preserve the normal form. Closure under
 964 vertical composition follows by stacking the two normal forms and collapsing the two independent
 965 A_i, ge, B_i blocks into a single block-diagonal matrix pair, as shown below:



966 For horizontal composition, the two normal forms are joined along the wire connecting B_1
 967 to A_2 ; applying Fourier-Motzkin Elimination to this pair recovers a fresh matrix-ge-comatrix
 968 triple, after which the two resulting runs of matrices, one in each direction, compose into a single
 969 matrix and a single comatrix, recovering the normal form:



970

□

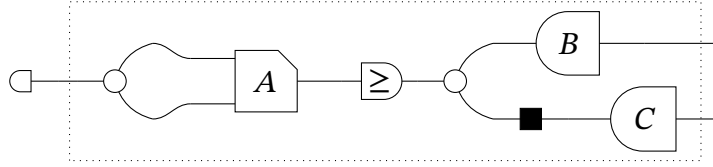
971

972

The following corollary reveals the geometric meaning of the normal form: the **GPA** part of every **GLP** diagram encodes the epigraph of the corresponding value function. This is the key

973 structural insight connecting the syntax to the semantic equivalence of Proposition 4.3.

974 **Corollary 4.1** (Epigraph Form). *Given a diagram $d \in \text{GLP}$ in normal form, it can always be*
 975 *rewritten as a diagram d' of the form:*



976 where the semantics of the dotted diagram is

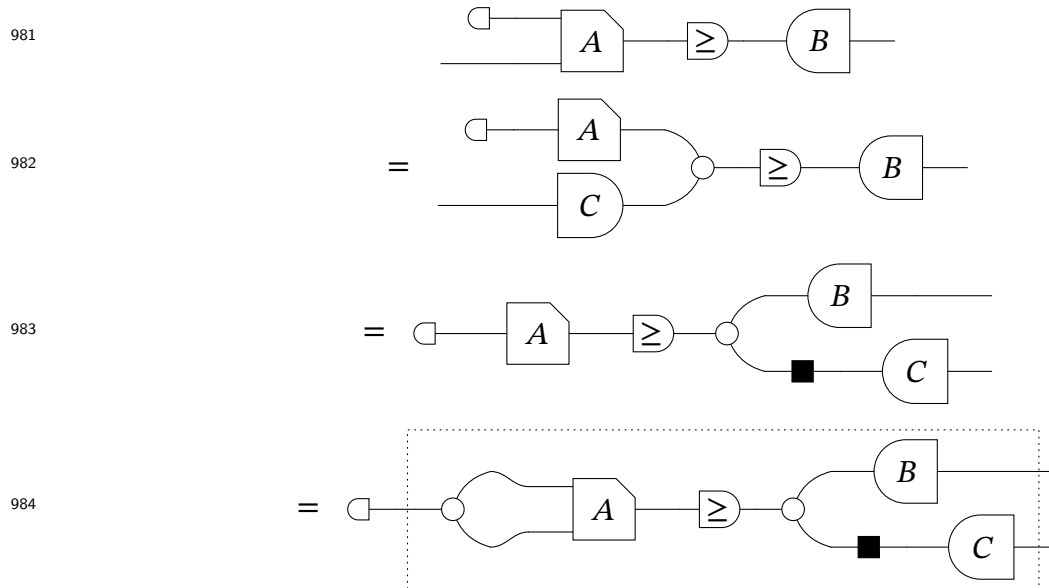
$$\left\{ (t, y) \mid \exists x, A'x + a \geq [B \ - C]y, t \geq x, y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right\}$$

977 . This is the epigraph of the original diagram d

$$\begin{aligned} [d](y_1, y_2) &= \inf_z z \quad \text{s.t. } Az + a \geq By_1 - Cy_2 \\ \text{epi}([d]) &= \{(t, y_1, y_2) \mid \exists z \leq t, Az + a \geq By_1 - Cy_2\} \end{aligned}$$

978 such that $A = [A' \ C]$. We call this the epigraph form of a linear program.

979 *Proof.* Starting from d in normal form, split the relation generator along $A = [A' \ C]$ and pull
 980 the resulting C -block through the $\geq$ generator using the antipode axiom:



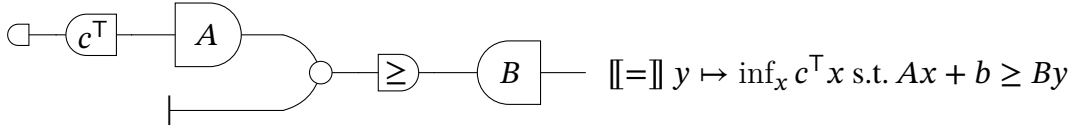
985 The resulting diagram is precisely the claimed form d' , with the dotted box exhibiting $\text{epi}([d])$
 986 as a GPA diagram. □

987 4.3 PRESENTATION RESULTS FOR LPREL

988 With the syntax and its normal form in place, we are now ready to prove the three presentation
989 properties for GLP.

990 **Lemma 4.1** (Expressivity). *For every $f : m \rightarrow n$ in $\mathbf{LPrel}$ there exists a Σ -term $s : m \rightarrow n$ in*
991 $\mathbf{Hom}_{\mathbf{FreeP}_{\mathbf{GLP}}}(m, n)$ *such that $[s] = f$.*

992 *Proof.* Any RHS Parametric linear program can be written as $\inf_z \{c^\top z \mid Ax + b \geq By\}$ therefore
993 is enough to show that we can always depict such a problem. The following diagram provides the
994 required syntactic representative:



995

□

996 **Lemma 4.2** (Soundness). *If $s = t$ in $\mathbf{FreeP}_{\mathbf{GLP}}$, then $[s] = [t]$ in $\mathbf{LPrel}$.*

997 *Proof.* All axioms inherited from **GPA** are already sound by the results of the previous section.
998 It suffices to verify that the new axioms involving the LP generator preserve the semantics, which
999 follows by direct computation:

$$\begin{aligned}
 (\text{linear.ge}) \quad & \Box \vdash \llbracket = \rrbracket (\bullet, y) \mapsto y = (\bullet, y) \mapsto \inf_{x \geq y} x \quad \llbracket = \rrbracket \quad \Box \vdash \geq \\
 (\text{lin.sum}) \quad & \Box \vdash \llbracket = \rrbracket (\bullet, (y_1, y_2)) \mapsto y_1 + y_2 \quad \llbracket = \rrbracket \quad \Box \vdash \bigcirc \\
 (\text{lin.zero}) \quad & \Box \vdash \llbracket = \rrbracket (\bullet, \bullet) \mapsto \inf_{x=0} x = 0 \quad \llbracket = \rrbracket \quad \Box \vdash \square
 \end{aligned}$$

1000

□

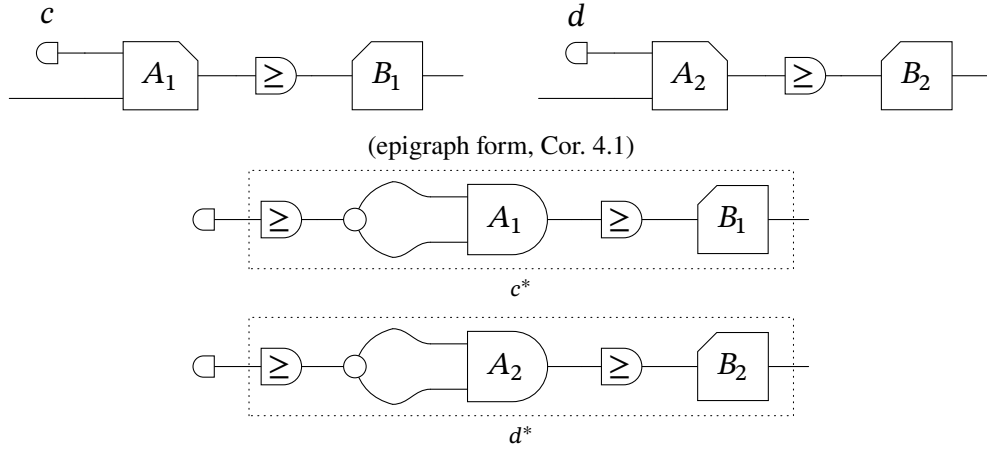
1001 The final result of the thesis is the completeness theorem for **GLP**. The proof brings together
1002 all the main tools developed throughout: the epigraph form of Corollary 4.1, the shared epigraph
1003 property of Proposition 4.3, the completeness of **GPA**, and the FME-based normal form.

1004 **Lemma 4.3** (Completeness). *If $[c] = [d]$ in $\mathbf{LPrel}$, then $c = d$ in $\mathbf{FreeP}_{\mathbf{GLP}}$.*

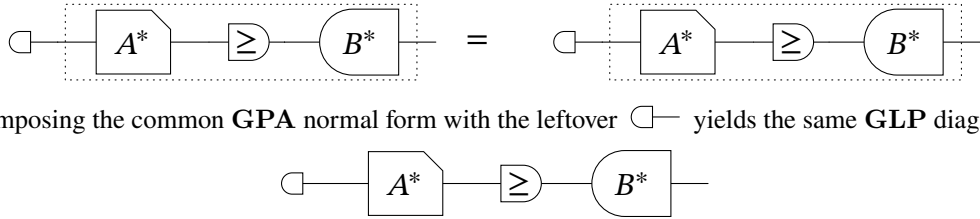
1005 *Proof.* Let $v_1 = [c]$ and $v_2 = [d]$. By assumption $v_1 = v_2$, so Proposition 4.3 gives $\text{epi}(v_1) =$
1006 $\text{epi}(v_2) =: E$.

1007 By Corollary 4.1, both c and d admit epigraph forms $c = \Box \vdash c^*$ and $d = \Box \vdash d^*$ with
1008 $[c^*] = [d^*] = E$ in $\mathbf{PolyRel}$. Since c^* and d^* are **GPA** diagrams with the same denotation,
1009 completeness of **GPA** gives $c^* = d^*$ in $\mathbf{FreeP}_{\mathbf{GPA}}$; in particular both reduce to the same **GPA**

1010 normal form. Composing this common diagram with the leftover $\square$ collapses c and d to the
 1011 same **GLP** diagram:



(the dotted boxes c^*, d^* are **GPA** diagrams with $[c^*] = [d^*] = E$; by **GPA** completeness both reduce to the same normal form)



1012 Therefore $c = \square$; $c^* = \square$; $d^* = d$. □

1013 As a consequence of the three lemmas above:

1014 **Theorem 4.2.** *LPRel is presented by $\text{FreeP}_{\text{GLP}}$.*

1015 *Proof.* By the preceding three lemmas. □

1016 Finally, we've built, through the addition of one generator and three axioms, an elegant and
 1017 complete diagrammatic theory for parametric linear programming. This result does unlock a new
 1018 algebraic and computational way to interact with LP problems. A linear program is ordinarily
 1019 presented as a function to be evaluated, an $x \mapsto c^\top x$ subject to constraints, solved from the outside
 1020 by an algorithm. Here it is instead a morphism, an object with an interface that composes with
 1021 others of its kind, and solving it, and eliminating one of its variables are all the same operation,
 1022 composition in **LPRel**.

1023 5 CONCLUSION

1024 We started with a remind of the quantale-valued relations and the quiet observation that
 1025 composing them is already a form of optimization, transitivity being its simplest instance. From
 1026 there, we recalled and refined the graphical theory of polyhedral relations, giving it a home
 1027 inside this same categorical hierarchy rather than treating it as a self-contained formalism. On
 1028 that foundation, we built **GLP**, a graphical algebra for parametric linear programs: a single new
 1029 generator for the linear objective, a handful of axioms, and a proof that these suffice, both for
 1030 completeness and for a normal form (Theorem 4.1).

1031 Two results are worth calling out on their own. First, **GLP** is *complete*: any two diagrams
 1032 with the same semantics can be rewritten into one another using only the axioms of the algebra,
 1033 so there is no gap between what the theory can prove equal and what is actually equal. Second,
 1034 every such diagram admits a *normal form* (Theorem 4.1), a canonical shape that every parametric
 1035 linear program reduces to, which is what makes the correspondence with value functions below
 1036 more than a loose analogy. This is one of the core properties of linear systems and thanks
 1037 to our Graphical Algebra is a syntactical operation in itself. As stated at the introduction, the
 1038 diagrammatic calculi makes the system properties evident in the syntax, instead of a higher level
 1039 result and this is the evidence.

1040 A linear program is now a fully composable object understood as a particular class of relations
 1041 such as the previous affine and polyhedral relations. Its diagrammatic syntax, once reduced to
 1042 normal form, coincides with the syntax of its own value function, so the diagram and the object
 1043 it computes are, in a precise sense, the same picture drawn twice.

1044 Just like that, other theories can be developed by enriching the here presented language.
 1045 Figure 5.1 lays out the resulting hierarchy of categorical structures that were developed by this
 1046 very recipe and what can actually be achieved in the end of the road.

1047 The **GLP** theory, in turn, has a natural extension to an even more expressive theory when
 1048 enriched with the gray arrow generator from **GQA** [Ste+24]. We believe that the class of
 1049 quadratic programs could be, just as was done in this work, fully axiomatized by such broader
 1050 **SMIT**. However, such studies are left for future works, along with other possible applications of
 1051 current results.

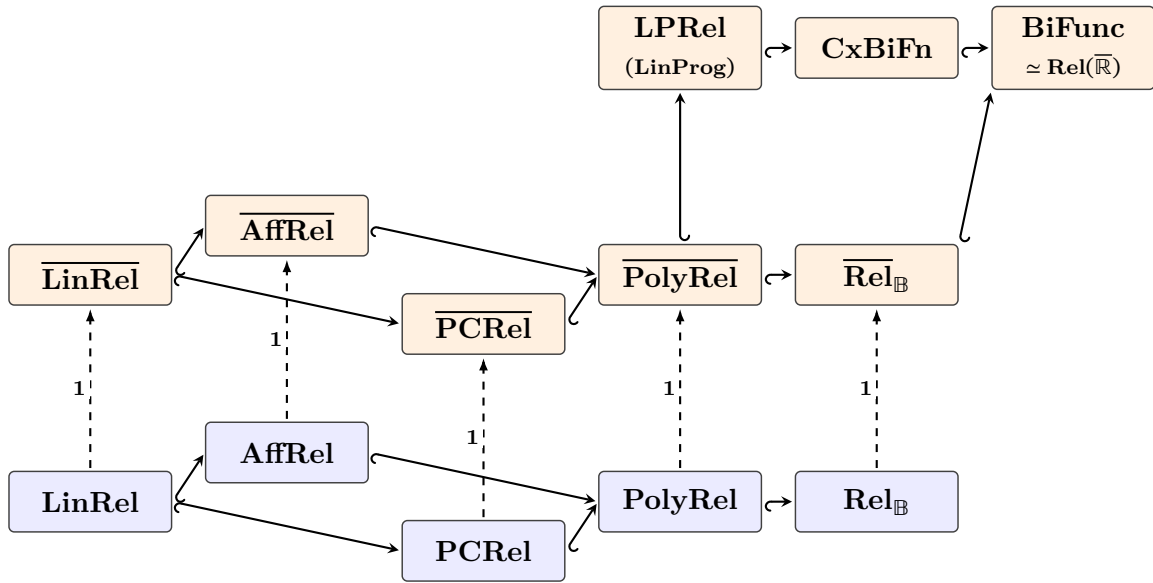


Figure 5.1 – Poset of semantic categories. Solid hooked arrows denote inclusion functors. Dashed arrows denote the indicator embedding $R \mapsto 1_R$, where $1_R = 0$ on R and $1_R = +\infty$ outside R .